

The Two-Time Moving-Defect Problem for Dispersing Billiards

Physical energy trees, tensor renewal, and radial Z -recovery

Working note

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Abstract

We isolate the estimate needed to differentiate an infinite correlation susceptibility under a genuine motion of the scatterers. A tempting bridge from the Demers–Liverani projective cone to forward standard families fails: the moving-face coefficient uses a one-sided trace on a transversal terminal arc, whereas the cone controls stable-leaf integrals. We give a local counterexample to any inference of such a trace bound from the published cone seminorms. The intrinsic positive problem is two-parameter equidistribution of a centered graph current under $T^{-1} \times T$. We encode it on a one-state countable Young quotient. An exact normal-cycle calculation gives the Anosov obstruction $\mathfrak{A} = -1$. Leclerc’s cohomological bracket identity identifies the physical temporal distance with the factor log-Jacobian four-point cocycle up to an exponentially small resolving tail. His all-scale escape modulo fixed-degree polynomials combines with the exact stable-holonomy Jacobian series to give a context–polynomial blow-up. We do not pass through an abstract countable QNL theorem. A gate-hitting estimate and a mass-preserving resolving buffer and a legal physical stopping cover feed Leclerc’s actual four-point cutting estimate into a conditional slope energy. A gap-mass recursion treats every physical cell and Jensen then gives the complete physical reduced-slope small-ball bound. The scale ledger separates the billiard remainder exponent from Leclerc’s cutting exponent.

For a reduced balanced phase $\psi_\xi = \xi_1 X + \xi_2 Y$, the exact identity $\psi'_\xi = X'(\xi_1 + \varepsilon_{\text{or}} \xi_2 e^S)$, $S = \log |Y'/X'|$, converts that theorem into a small-ball estimate for the *value* of the log slope. Smooth cutoffs in S isolate the resonant set; on its complement an elementary $C^{1+\alpha}$ fractional nonstationary-phase lemma is applied directly to the physical unstable conditional. Its scaled density and mark moment are derived from collision density, holonomy, and chart Jacobians, and its exponent budget is explicit. A dominant-coordinate decomposition removes contracted-coordinate Fourier modes before oscillatory analysis. No Wronskian, second quotient derivative, or connected-component phase partition is used. Two global marginal projections act in a direct sum of graph, rank-one, and regular-product currents before graph disintegration. The full A – T – B – C renewal identity is tensorized on explicitly defined strong/weak current spaces, with observables inserted only in the final dual pairing and all other poles excluded on a common spectral disk. A two-chart coarea disintegration and a forward/reverse stopping tree for each curved radial face give the two mixed recovery quadrants without dividing by a vanishing coordinate derivative. A negative short-curve moment gives the required boundary-functional recursion. Exponential return tails and the one-time grazing-strip estimate yield a joint return–maximal-grazing moment by a union bound followed by Hölder, without independence. Finally, for each finite return cutoff the parameter atlas is indexed by finite physical cut histories, not fibers of an infinite sign sequence; finite-depth generic face separation and a scalar conditional marked moment on the actual label filtration remove the collision and

grazing cutoffs uniformly. Coarea-current convergence replaces componentwise common atlases at fixed depth, central decay is proved before recovery summation, and the common bi-recovery moment supplies dominated convergence. The regular two-dimensional defect is treated in the same tensor current space. Under the primitive marked-quotient, growth, clean cycle, and coarea hypotheses listed below this yields CM2 for generic clean circular tables.

1 The exact target

Throughout, $\{T_s : |s| < s_0\}$ is a C^3 compact family of finite-horizon planar dispersing billiard collision maps obtained by a prescribed radial motion of circular scatterers and transported to one reference collision space M . The table at $s = 0$ contains the unobstructed normal cycle used in Proposition 6.8. We consider only the complete *first* moving-table defect: regular Jacobian/density derivatives and all same-label physical one-sided faces are assembled before centering. The tests f, g belong to $C^r(M)$, with r above the Fourier summability threshold, and $\mu_0(g) = 0$. Second shape defects and unassembled point atoms are outside the statement.

Let P_s be the Markov-normalized transfer operator on a fixed collision fiber, $P_s \mathbf{1} = \mathbf{1}$, let

$$\Pi h = \mu_0(h) \mathbf{1}, \quad Q = P_0 - \Pi,$$

and let K_V denote the complete first moving-table defect after all regular terms, same-label physical faces, and grazing blocks have been assembled. The two-time estimate needed for a prescribed centered source g is

$$|\langle Q^n K_V Q^m g, f \rangle| \leq C \tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r}, \quad m, n \geq 0, \quad (1)$$

for some $\tau, \vartheta < 1$. The same estimate, uniformly for $s^{-1}(P_s - P_0)$, is required before a parameter limit can be exchanged with the two time sums.

The operator must be centered as a whole:

$$K_V = \Pi^\perp \partial_s P_s|_{s=0} \Pi^\perp. \quad (2)$$

Centering separate faces is neither invariant nor sufficient. Without (2), rank-one terms generally give an additive bound $O(\tau^m + \vartheta^n)$, whose double sum diverges.

2 Why the direct projective-cone bridge fails

The projective cone for dispersing billiards controls integrals of a density against positive dynamically Hölder tests on stable curves, short-leaf mass, and Hölder comparison of such integrals between nearby stable curves. A moving inverse-branch boundary instead produces, before physical assembly, a coefficient of the form

$$h \longmapsto w(h|_{I^+} - h|_{I^-}) \delta_\Gamma, \quad (3)$$

where $I^\pm$ are one-sided terminal preimage arcs. In the radial collision atlas these arcs land on the grazing seam and are transversal to the stable test foliation. Thus (3) asks for a transverse trace, not a stable-leaf average.

The following elementary local model rules out the missing implication.

Proposition 2.1 (Stable-leaf control does not dominate transverse trace). *Let $R = (-2, 2)^2$, let the local stable leaves be the horizontal segments $W_y = (-2, 2) \times \{y\}$, and let the transversal face be $I = \{0\} \times (-1, 1)$. Fix $0 < q < 1$ and $0 < \gamma < 1$. There are smooth functions u_ε supported in an ε -tube about I such that*

$$u_\varepsilon|_I = b \neq 0,$$

while, uniformly over bounded Hölder test functions ψ on stable subsegments $W \subset W_y$,

$$\left| \int_W u_\varepsilon \psi \, dm_W \right| \leq C \min\{|W|, \varepsilon\} \|\psi\|_\infty, \quad (4)$$

$$\sup_{0 < |W| \leq 1} |W|^{-q} \left| \int_W u_\varepsilon \psi \, dm_W \right| \leq C \varepsilon^{1-q} \|\psi\|_\infty, \quad (5)$$

and the Hölder comparison of these integrals between nearby parallel stable leaves is $O(\varepsilon)$. Consequently no norm or order estimate using only the stable-leaf quantities in (4)–(5) and their transverse Hölder comparisons can bound $\|u|_I\|_{C^0}$.

Proof. Choose $\phi \in C_c^\infty((-1, 1))$ with $\phi(0) = 1$ and $b \in C_c^\infty((-1, 1))$, $b \neq 0$, and set

$$u_\varepsilon(x, y) = \phi(x/\varepsilon)b(y).$$

The trace assertion is immediate. A horizontal subsegment meets the support in length at most $C \min\{|W|, \varepsilon\}$, proving (4). If $|W| \leq \varepsilon$, multiplication by $|W|^{-q}$ gives at most $C \varepsilon^{1-q}$; if $|W| \geq \varepsilon$, it gives at most $C \varepsilon |W|^{-q} \leq C \varepsilon^{1-q}$, proving (5). Replacing y by y' changes the integral by at most $C \varepsilon |y - y'| \|b\|_{C^1} \|\psi\|_{C^0}$, with the usual additional Hölder test-comparison term. The trace is independent of ε , so the asserted domination is impossible. $\square$

Corollary 2.2 (Failure of the proposed positive cone operator). *The local face map (3) is not known to be positive for the Demers–Liverani cone order, and its boundedness cannot be inferred from projective contraction. In particular, the implication*

$$\Theta(P_0^m f_+, P_0^m f_-) \leq C \chi^m \implies \|K_V P_0^m (f_+ - f_-)\|_Y \leq C' \chi^m$$

is invalid without an additional transverse-trace theorem.

Proof. The cone order is tested through stable-leaf integrals. The face coefficient contains precisely the trace excluded by Proposition 2.1. Projective contraction therefore supplies no order comparison for the two output face measures. $\square$

This is a go/no-go conclusion, not merely a missing estimate: the direct “past cone $\rightarrow$ face $\rightarrow$ future family” proof must be abandoned unless physical assembly first converts the terminal trace into an integral functional controlled by the cone. The existing adjoint dyadic bound does not perform that conversion.

3 A sufficient two-sided bridge

The complete defect is a kernel located at the event time. It must be controlled from the past and the future simultaneously. The following criterion is an exact sufficient condition, but not a characterization of CM2.

Definition 3.1 (Centered nuclear recovery bridge). *Let X_- be a Banach space of centered past observables and let Y_+ be a Banach space of centered signed measured unstable families. A centered operator $K : X_- \rightarrow Y_+$ has a nuclear recovery presentation if*

$$K = \sum_{j \geq 1} \ell_j \otimes \eta_j, \quad \sum_{j \geq 1} \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+} < \infty, \quad (6)$$

where $\ell_j(\mathbf{1}) = 0$ and $\eta_j(\mathbf{1}) = 0$ after the complete regular and physical-face assembly.

Theorem 3.2 (Two-sided bridge implies CM2). *Assume*

$$\|Q^m g\|_{X_-} \leq C_- \tau^m \|g\|_{X_-}, \quad (7)$$

$$|\eta(\mathcal{U}_0^n f)| \leq C_+ \vartheta^n \|\eta\|_{Y_+} \|f\|_{C^1}, \quad (8)$$

for centered g and centered η . If K has the presentation (6), then

$$|(KQ^m g)(\mathcal{U}_0^n f)| \leq C_- C_+ \mathfrak{N}(K) \tau^m \vartheta^n \|g\|_{X_-} \|f\|_{C^1},$$

where $\mathfrak{N}(K) = \sum_j \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+}$.

Proof. Using (6), (7), and (8),

$$\begin{aligned} |(KQ^m g)(\mathcal{U}_0^n f)| &\leq \sum_j |\ell_j(Q^m g)| |\eta_j(\mathcal{U}_0^n f)| \\ &\leq C_- C_+ \tau^m \vartheta^n \|g\|_{X_-} \|f\|_{C^1} \sum_j \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+}. \end{aligned}$$

□

Remark 3.3. *The theorem makes clear why a usual functional correlation estimate with error $O(\tau^m + \vartheta^n)$ is insufficient. CM2 requires a centered operator ideal estimate at the event time, not just mixing on either side. Wormell's specialized two-time conditional-mixing conjecture is one way to encode precisely this product structure.*

Remark 3.4 (The nuclear criterion is generally too strong). *A physical face current is supported on a graph in a stable-unstable product chart. Its kernel is therefore analogous to a pullback or a Fourier integral operator, not to a smoothing operator. Even after double centering, a graph delta need not be nuclear between natural past and future spaces. Thus failure of (6) or of the projective-tensor condition below does not imply failure of CM2. Product decay may instead arise dynamically, from oscillation and transversality after the graph is acted on by both time directions.*

4 Recovery-time weighted future space

For a measured unstable component (W, ρ) let $R(W)$ be the first time at which its canonical forward subdivision is a proper standard family. For $\omega > 0$, define schematically

$$\|\eta\|_{Y_+} := \inf \sum_j |a_j| e^{\omega R(W_j)} (1 + \text{Reg}(\rho_j)), \quad \eta = \sum_j a_j (W_j, \rho_j). \quad (9)$$

The infimum also allows the positive and negative Jordan parts to be split by recovery rank. Standard-family coupling and recovery give (8) once the constants in (9) are chosen within the recovery tail.

For a radial grazing block b , the physical flux factorization gives a mass bound $M_b \leq C2^{-b}$. If its geometric deterioration is polynomial in 2^b , the recovery lemmas give

$$R_b \leq C_0 + C_1 b.$$

Hence

$$\sum_{b \geq 0} M_b e^{\omega R_b} < \infty \quad \text{whenever} \quad \omega C_1 < \log 2. \quad (10)$$

Thus initial finite Z is unnecessary. What remains is not future recovery, but the past functional ℓ_j in (6).

4.1 Recovery windows and the actual summability target

For differentiability of the susceptibility, a uniform pointwise CM2 bound for every grazing block is stronger than necessary. The required conclusion is absolute summability of the complete block series. This distinction removes an artificial exponential loss in the grazing rank.

Proposition 4.1 (Recovery-window summability). *Let $B_b(m, n)$ be the scalar pairing produced by block b at past and future times m, n . Suppose that there are masses M_b , recovery times R_b^-, R_b^+ , constants $C < \infty$, and $0 < \tau, \vartheta < 1$ such that*

$$|B_b(m, n)| \leq C M_b \tau^{(m-R_b^-)_+} \vartheta^{(n-R_b^+)_+}, \quad m, n \geq 0. \quad (11)$$

Then

$$\sum_{b, m, n} |B_b(m, n)| \leq C' \sum_b M_b (1 + R_b^-)(1 + R_b^+). \quad (12)$$

In particular, if $M_b \leq C2^{-\alpha b}$ for some $\alpha > 0$ and $R_b^\pm \leq C_0 + C_1 b$, then the full threefold series converges. The same conclusion holds uniformly for parameter difference quotients if the constants and block masses are uniform.

The deterministic recovery times may be replaced by a countable refinement $B_b = \sum_j B_{b,j}$ with masses $M_{b,j}$ and times $R_{b,j}^\pm$, provided

$$\sum_j M_{b,j} (1 + R_{b,j}^-)(1 + R_{b,j}^+) \leq C M_b (1 + b)^2. \quad (13)$$

It is enough for the two-sided recovery tail, after its deterministic $O(b)$ entrance time, to be exponential.

Proof. For every $R \geq 0$ and $0 < q < 1$,

$$\sum_{k \geq 0} q^{(k-R)_+} \leq R + 1 + \frac{1}{1-q}.$$

Sum (11) first in m, n and then in b . Exponential block mass dominates the quadratic recovery window. Applying the same calculation to every refined piece gives (13). An exponential tail for $R_{b,j}^- + R_{b,j}^+ - O(b)$ has uniformly bounded polynomial moments, proving the last assertion. $\square$

Lemma 4.2 (Two marginal growth tails imply bi-recovery). *Let a block of total variation M_b be cut by the common refinement of its past and future recovery trees, with refined masses $M_{b,j}$ and recovery times $R_{b,j}^\pm$. Assume cutting preserves total variation and, for some $r_b \leq C_0 + C_1 b$, $0 < \rho < 1$,*

$$\sum_{j: R_{b,j}^\pm > r_b + t} M_{b,j} \leq C M_b \rho^t, \quad t \geq 0,$$

for each sign. Then (13) holds.

Proof. If $R_{b,j}^- + R_{b,j}^+ > 2r_b + t$, at least one marginal time is larger than $r_b + t/2$. The union bound therefore gives an exponential tail for the sum of the two recovery times under the normalized weights $M_{b,j}/M_b$. Its second moment is $O((1+r_b)^2)$. Since $(1+R^-)(1+R^+) \leq (1+R^-+R^+)^2$, the claim follows. $\square$

Proposition 4.3 (Boundary-functional reduction of proper bi-recovery). *Assume the grouped face current satisfies the two-sided moment*

$$\sum_{\lambda} M_{\lambda} [(1 + Z_q(\mathcal{G}_{\lambda}^-))(1 + Z_q(\mathcal{G}_{\lambda}^+))]^{\epsilon} < \infty. \quad (14)$$

Then logarithmic recovery implies

$$\sum_{\lambda} M_{\lambda} (1 + R_{\lambda}^-)(1 + R_{\lambda}^+) < \infty.$$

Proof. Use $R_{\lambda}^\pm \leq a_1 \log(1 + Z_q(\mathcal{G}_{\lambda}^\pm)) + a_2$. Every polynomial in the two recovery times is dominated by a smaller positive power of the product in (14). $\square$

Corollary 4.4 (Clean radial blocks satisfy bi-recovery). *For the grouped first radial face current, Lemma 6.40 and Proposition 6.41 prove (14) uniformly. Hence all recovery windows are summable after the central estimate.*

Proof. The initial radial negative moment is (89); recursive cuts obey (85). Apply Proposition 4.3 and then Proposition 4.1. $\square$

For a general graph current the relevant geometric input is therefore a *proper bi-recovery* statement, not an estimate of a Fourier constant on the original grazing curve. For the clean first radial current this input is Corollary 4.4. A central two-time theorem then gives (11). The fully unrecovered rectangle is handled by total variation, the two mixed rectangles by the corresponding recovered one-sided estimate, and the fully recovered rectangle by the product bound. The growth-lemma tail sums these windows through (13).

Consequently even the generic $M_b = O(2^{-b})$ assembly margin is sufficient once proper bi-recovery is proved; the stronger first-radial $O(2^{-(3-\epsilon_*)b})$ margin gives additional room but is no longer needed to absorb an exponential Fourier constant in b . This decouples the grazing rank from the return-rank truncation in Proposition 6.4.

5 Billiard-specific block problem

For each genuine physical block b , the complete difference quotient must be written, after artificial-face cancellation, as

$$K_{V,b} = \sum_{k \geq 1} \ell_{b,k} \otimes \eta_{b,k}, \quad (15)$$

with

$$\sum_{b,k} \|\ell_{b,k}\|_{X_-^*} \|\eta_{b,k}\|_{Y_+} < \infty. \quad (16)$$

The desired estimate follows from Theorem 3.2. Three sufficient mechanisms for proving (16) are:

- (i) a reverse-time conditional-mixing theorem for the terminal slice, furnishing X_-^* directly;
- (ii) a two-sided Young-tower lift of the complete centered defect, with an event kernel that is nuclear between the past and future tower norms;
- (iii) a smoothing/nuclear decomposition of the terminal restriction, with a quantitative bunching inequality strong enough to sum the smoothing error over the homogeneity blocks.

The projective-cone contraction alone supplies none of these three inputs. These mechanisms are not exhaustive because a non-nuclear graph current can still equidistribute under the two-sided dynamics.

5.1 A two-sided tower formulation

The natural extension gives a sharper version of alternative (ii). Let Δ be a two-sided Young tower for the fixed billiard, with past and future cylinder filtrations $\mathcal{F}_i^-$ and $\mathcal{F}_j^+$. Denote the associated martingale differences by

$$D_i^- = \mathbb{E}(\cdot \mid \mathcal{F}_i^-) - \mathbb{E}(\cdot \mid \mathcal{F}_{i-1}^-), \quad D_j^+ = \mathbb{E}(\cdot \mid \mathcal{F}_j^+) - \mathbb{E}(\cdot \mid \mathcal{F}_{j-1}^+).$$

A lifted event kernel $\widehat{K}$ has summable mixed cylinder variation if

$$\sum_{i,j \geq 1} \|D_i^- D_j^+ \widehat{K}\|_{\mathcal{B}_-^* \widehat{\otimes}_\pi \mathcal{B}_+} < \infty. \quad (17)$$

The indices start at one: complete double centering removes the two axis terms $D_i^- D_0^+ \widehat{K}$ and $D_0^- D_j^+ \widehat{K}$.

Proposition 5.1 (Mixed-cylinder criterion). *Suppose the two-sided tower has exponential tails and the past and future transfer actions contract the centered martingale blocks at rates τ^m and ϑ^n . If the complete lifted moving defect satisfies (17), then it has a nuclear recovery presentation (6) and hence satisfies CM2.*

Proof. Expand

$$\widehat{K} = \sum_{i,j \geq 1} D_i^- D_j^+ \widehat{K}$$

in the projective tensor product. Each summand is a norm-convergent sum of rank-one past/future tensors by the definition of $\widehat{\otimes}_\pi$. Summability in (17) therefore gives (6). Shifting m steps into the past and n steps into the future contracts the two centered tensor factors separately. Apply Theorem 3.2. $\square$

This criterion identifies one strong geometric issue. A smooth two-dimensional density on a tower rectangle has summable mixed cylinder variation. A face current is instead supported on a one-dimensional graph in the local stable-unstable product chart. Its graph delta is not automatically in the projective tensor product. Proving (17) is therefore a genuine conditional-mixing theorem, not routine tower bookkeeping. The finite-strip central face supplies uniform transversality and one-sided recovery, but these facts alone do not prove the mixed tensor estimate.

6 The intrinsic dynamic graph-current formulation

Let $\gamma_b : I_b \rightarrow M$ parametrize one side of a complete physical face in block b , let $F_b\gamma_b$ denote its assembled terminal image, and let w_b be the signed coarea weight after artificial-face cancellation. The face contribution to CM2 has the model form

$$B_{m,n}^{(b)}(g, f) = \int_{I_b} g(T^{-m}\gamma_b(t))f(T^n F_b\gamma_b(t))w_b(t) dt. \quad (18)$$

Equivalently, if $J_{V,b}$ is the graph current on $M \times M$ induced by $t \mapsto (\gamma_b(t), F_b\gamma_b(t))$, then

$$B_{m,n}^{(b)}(g, f) = ((T^{-m} \times T^n)_* J_{V,b})(g \otimes f). \quad (19)$$

Complete double centering removes the two marginals of $J_{V,b}$. The intrinsic CM2 statement is therefore two-parameter equidistribution of a centered graph current under $T^{-1} \times T$ at a product rate.

This formulation explains both the difficulty and the opportunity. A graph curve is one-dimensional, whereas the unstable bundle of the product map is two-dimensional, so ordinary standard-family coupling does not apply directly. On the other hand, the current need not be statically nuclear: hyperbolic stretching sends its Fourier frequencies to exponentially large scales, and polynomial Fourier decay of the induced slice is already enough to produce exponential decay in collision time.

Proposition 6.1 (Fourier reduction on one induced rectangle). *Assume that an induced return map has a stable–unstable product chart in which the centered face slice induces a finite signed measure ν_b on the contracting coordinate. Suppose that, uniformly over admissible inverse words and over the relevant C^2 phase maps F ,*

$$|\widehat{F_*\nu_b}(\xi)| \leq C_b(1 + |\xi|)^{-\eta} \quad (20)$$

for some $\eta > 0$. Suppose also that, after excluding phase cancellation, the expanding frequencies generated by m past returns and n future returns are bounded below by

$$c \max\{\Lambda_-^m, \Lambda_+^n\},$$

apart from cylinder pieces whose total centered mass is bounded by $C\tau_0^m\vartheta_0^n$. Then, for sufficiently regular centered cylinder observables,

$$|B_{m,n}^{(b)}(g, f)| \leq CC_b \tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r} \quad (21)$$

for some $\tau, \vartheta < 1$.

Proof. Decompose g and f into Fourier modes in the product chart and split the sum into low and high frequencies. Centering and cylinder mixing control the low-frequency part. In the high-frequency part, hyperbolicity converts the two return depths into the lower frequency scale $c \max\{\Lambda_-^m, \Lambda_+^n\}$. Since

$$\max\{\Lambda_-^m, \Lambda_+^n\} \geq (\Lambda_-^m \Lambda_+^n)^{1/2},$$

(20) gives the product factor $\Lambda_-^{-\eta m/2} \Lambda_+^{-\eta n/2}$. Summability of the Fourier coefficients of C^r tests and the exceptional-cylinder bound complete the estimate. The loss of one half in each exponent is harmless. $\square$

This preliminary proposition is not used as an assumption in the final proof. Uniform (20) is supplied below by the cohomological local slope-sublevel theorem, the dominant-coordinate reduction, and the LPS nonstationary-phase estimate. One-sided mixing alone would not supply it.

6.1 A three-region reduction

Fourier analysis is not needed uniformly over the whole (m, n) quadrant. One-sided standard-family recovery gives a forward estimate with a past regularity cost, and reversibility gives its backward analogue. The following elementary lemma isolates the remaining diagonal band.

Lemma 6.2 (Off-diagonal coupling plus diagonal oscillation). *Suppose*

$$|B_{m,n}| \leq C e^{\beta_- m - \alpha_+ n}, \quad (22)$$

$$|B_{m,n}| \leq C e^{-\alpha_- m + \beta_+ n}, \quad (23)$$

with all four exponents positive. Choose $A > \beta_-/\alpha_+$ and $D > \beta_+/\alpha_-$. Suppose in addition that

$$|B_{m,n}| \leq C e^{-\gamma \max\{m,n\}}$$

whenever $D^{-1}m < n < Am$. Then there are $\tau, \vartheta \in (0, 1)$ such that $|B_{m,n}| \leq C' \tau^m \vartheta^n$ for all m, n .

Proof. If $n \geq Am$, (22) is bounded by $e^{-c_+(m+n)}$ for some $c_+ > 0$. If $m \geq Dn$, use (23). The remaining indices lie in the stated diagonal band, where $\max\{m, n\} \geq (m+n)/2$. Taking the minimum of the three positive exponents proves the claim. $\square$

For billiards, (22) follows from signed standard-family coupling after cutting the past-weighted density into homogeneous pieces: the growth lemma gives the coupling factor $e^{-\alpha_+ n}$, while differentiating the m -step past weight costs at most $e^{\beta_- m}$. The marked-amplitude moment sums the homogeneous pieces. Applying the same argument to the reversible collision map gives (23). All constants are uniform on the fixed compact atlas; recovered grazing pieces enter only after their recorded recovery times; these are the signed standard-family forms of the usual growth and coupling estimates [18, 2]. Consequently the genuinely new slope/Fourier theorem is needed only for $m \asymp n$.

This also corrects the naive ambient phase test. The graph $t \mapsto (T^{-m}\gamma(t), T^n F\gamma(t))$ is one-dimensional in a four-dimensional product. Its first two derivatives cannot be uniformly transverse to every Fourier covector. In the diagonal band one must first discard the contracted coordinates and pass to the dominant past-stable and future-unstable coordinates. The resulting two-dimensional plane curve is governed by the stable-unstable temporal-distance/holonomy function. A finite-difference slope-value QNL is appropriate for this reduced phase. If one insists on working in the full four-dimensional chart, a finite-type condition involving derivatives through order at least four is required.

6.2 A verified radial nonlinearity seed

The first radial face is not an abstract graph. For two circular scatterers with radii R_j, R_i and centre distance $d > R_j + R_i$, put the source centre at the origin, the target centre at $(d, 0)$, and parametrize the source boundary by

$$q_j(\alpha) = R_j(\cos \alpha, \sin \alpha).$$

Let $r(\alpha) = |(d, 0) - q_j(\alpha)|$ and $\beta(\alpha) = \arg((d, 0) - q_j(\alpha))$. The two directions tangent to the target circle are

$$\theta_\sigma(\alpha) = \beta(\alpha) + \sigma \arcsin \frac{R_i}{r(\alpha)}, \quad \sigma \in \{-1, 1\}.$$

Thus the radial physical face in Birkhoff coordinates is locally $\varphi_\sigma(\alpha) = \theta_\sigma(\alpha) - \alpha$.

Lemma 6.3 (The circular tangency face is genuinely curved). *At the collinear source point $\alpha = 0$, write $r_0 = d - R_j$. Then*

$$\varphi''_\sigma(0) = -\sigma \frac{R_i d R_j}{r_0^2 \sqrt{r_0^2 - R_i^2}} \neq 0. \quad (24)$$

Moreover the angular coarea derivative of

$$G = |((d, 0) - q_j) \times v_\theta|^2 - R_i^2$$

is nonzero there. Hence each tangency choice gives a smooth, non-affine central face germ with a smooth coarea density.

Proof. Direct differentiation gives

$$r'(0) = 0, \quad r''(0) = \frac{dR_j}{r_0}, \quad \beta''(0) = 0.$$

Since

$$\frac{d}{dr} \arcsin \frac{R_i}{r} = -\frac{R_i}{r \sqrt{r^2 - R_i^2}},$$

equation (24) follows. On the tangency face, $|((d, 0) - q_j) \times v_\theta| = R_i$ while the longitudinal component has magnitude $\sqrt{r_0^2 - R_i^2}$; therefore $|\partial_\theta G| = 2R_i \sqrt{r_0^2 - R_i^2} > 0$. $\square$

This proves a useful but limited fact: the radial face supplies a concrete one-step nonlinearity seed and no Baker-type affine resonance is forced by the circular geometry. It does not prove uniform quantitative nonlinearity after arbitrary past and future words. Near the diagonal where the two terms in the reduced phase

$$\Phi_{m,n,\xi,\zeta}(t) = \xi \cdot T^{-m} \gamma(t) + \zeta \cdot T^n F \gamma(t) \quad (25)$$

have comparable size, their derivatives can cancel; this is the only time regime in which a genuine Dolgopyat/sum-product argument is needed. Off that diagonal one time direction dominates and ordinary nonstationary phase applies.

6.3 A countable-alphabet truncation lemma

The latest smooth result closest to the required mechanism is Leclerc's Fourier-decay theorem for equilibrium states of nonlinear, area-preserving smooth Axiom-A surface diffeomorphisms. Its quantitative nonlinearity condition is expressed through a temporal-distance function and is generic, with an explicit periodic-point cocycle test. A billiard return scheme is piecewise smooth with a countable alphabet, so that theorem is not directly applicable. The countable alphabet itself, however, is not the fundamental obstruction.

Proposition 6.4 (Exponential-tail reduction to uniform finite truncations). *Let an induced symbolic system have a return/singularity rank R with*

$$\mu\{R > L\} \leq C e^{-aL}.$$

Suppose a Fourier proof at frequency $|\xi|$ uses at most $N(\xi) \leq c_0 \log(2 + |\xi|)$ successive symbols. Assume that the subsystem with all ranks at most L satisfies, uniformly in the truncation,

$$\left| \int e^{i\xi\psi} \chi d\mu_L \right| \leq C e^{cL} |\xi|^{-\eta}.$$

Then the full countable system has polynomial Fourier decay, provided $a, \eta > 0$ and the finite-truncation constants have the displayed exponential bound.

Proof. The mass of orbit words which see a rank larger than L is at most $CN(\xi)e^{-aL}$. On the complement apply the finite-truncation estimate. Choose $L = \delta \log |\xi|$ with $0 < \delta < \eta/c$ (with no upper restriction if $c = 0$). The good contribution is $O(|\xi|^{-\eta+c\delta})$ and the bad contribution is $O(\log |\xi| |\xi|^{-a\delta})$. Both have a positive power saving. $\square$

Thus this fallback would require a *uniform* singular version of Leclerc's quantitative-nonlinearity estimate on finite homogeneity/return truncations. Exponential tower tails could then remove the truncation, but the truncation-uniform constants are not supplied by that route. The direct gate-resolved slope argument below bypasses this fallback entirely.

6.4 Uniform QNL requires anchored big images

Inspection of the QNL proof shows that a single nonzero periodic cocycle is not, by itself, a uniform truncation theorem. The power exponent also depends on a cutting constant: at every scale one removes a child interval carrying at least an η_{cut} fraction of its parent's conditional mass. For an arbitrary exhaustion by finite subshifts, η_{cut} may tend to zero; then the QNL exponent tends to zero and Proposition 6.4 is useless.

The billiard inducing scheme has a possible cure. A Gibbs–Markov return map has a finite big-images/preimages core, and on a compact central magnet the SRB conditional densities on homogeneous unstable leaves are bounded above and below. The truncations must therefore be *anchored*: retain a fixed finite family of full-image connector branches, including the nonlinear periodic witness, and add all branches up to rank L . The desired uniform cutting statement is

$$\nu_{x,L}(I_{\text{child}}) \geq \eta_0 \nu_{x,L}(I_{\text{parent}}), \quad \eta_0 > 0 \quad (26)$$

with η_0 independent of x, L , for at least one QNL-separated child at each scale. If (26) and a uniform UNI constant $\kappa_0 > 0$ hold, Leclerc's tree argument gives a QNL exponent $\Gamma > 0$ independent of L ; only the multiplicative constant is allowed to grow like e^{cL} .

Rühr's pressure inequalities quantify weak convergence of Gibbs measures on countable Markov shifts to equilibrium measures on finite subsystems by the square root of the pressure gap. They support the truncation step, but do not prove (26) or uniform UNI. Strong positive recurrence/BIP controls the symbolic escape; the billiard geometry must still provide the nonlinear separated child.

For the collision SRB measure on a compact homogeneous magnet, the unstable conditional is better than a general fractal Gibbs conditional: in arclength coordinates its density is bounded above and below. In that case the measure-theoretic part of the QNL argument is elementary.

Lemma 6.5 (Uniform interval UNI implies power non-concentration). *Let ν_L be measures on an interval whose densities satisfy $0 < c_- \leq d\nu_L/dt \leq c_+ < \infty$, uniformly in L . Let X_L be a family of functions with the following scale-invariant UNI property: for every interval J of length $\sigma < \sigma_0$ and every polynomial P in a fixed finite-dimensional class, there is a subinterval $J' \subset J$ with $|J'| \geq \kappa|J|$ such that*

$$|X_L(t) - P(t)| \geq \kappa|J| \quad (t \in J'),$$

where $\kappa > 0$ is independent of J, P, L . Then for every fixed $\alpha > 0$ there are $C, \Gamma > 0$, independent of L , such that

$$\nu_L\{t \in J : |X_L(t) - P(t)| \leq |J|^{1+\alpha}\} \leq C|J|^\Gamma \nu_L(J).$$

Proof. Partition every surviving interval into a bounded number of comparable children and discard a child contained in the UNI interval J' . The density bounds imply that at least a fixed fraction $\eta_0 > 0$ of the conditional mass is discarded at each generation. Iterate until the child scale is comparable to $|J|^{1+\alpha}$. The number of generations is $c_\alpha |\log |J|| + O(1)$, so the surviving mass is at most $(1 - \eta_0)^{c_\alpha |\log |J||} \nu_L(J) \leq C|J|^\Gamma \nu_L(J)$. $\square$

Thus for the central homogeneous face the principal task can be reduced further: establish a uniform geometric UNI statement for the stable–unstable temporal-distance function. Finite-subsystem approximation is then needed for long singular itineraries and difference quotients, but not to manufacture the QNL exponent on the central magnet.

6.5 A fixed two-branch UNI certificate

Bertozzi's 2026 countable-Markov exponential-mixing construction provides a useful template: after sufficient inducing, two fixed inverse branches are repeated and the derivative of their Birkhoff-roof difference is bounded away from zero uniformly in the iteration depth. The same device can be formulated for the billiard temporal-distance increment.

Let h_0, h_1 be two regular inverse return branches on the central magnet, with a common contraction bound $\lambda < 1$, and let τ denote the reduced stable–unstable phase increment (for the geometric application, $\tau = \log J^u F$ after the stable coboundary normalization). Define

$$\Psi_N(x) = \sum_{k=1}^N [\tau(h_0^k x) - \tau(h_1^k x)], \quad \Psi_\infty = \lim_{N \rightarrow \infty} \Psi_N. \quad (27)$$

Proposition 6.6 (Anchored two-branch certificate). *Suppose the series in (27) converges in C^1 and*

$$\inf_{x \in I} |\Psi'_\infty(x)| \geq c_0 > 0$$

on a central unstable interval I . Then, for all sufficiently large N ,

$$\inf_{x \in I} |\Psi'_N(x)| \geq c_0/2.$$

If the two branches and their fixed connector branches are retained in every anchored truncation, this UNI constant is independent of the truncation rank.

Proof. C^1 convergence makes $\|\Psi'_N - \Psi'_\infty\|_\infty < c_0/2$ for all large N . The second assertion follows because the calculation uses only the anchored branches, not the additional symbols admitted by a larger truncation. $\square$

Corollary 6.7 (First-term domination after acceleration). *Assume $\|Dh_i\| \leq \lambda < 1$, $\|D\tau\|_\infty \leq M$, and*

$$\inf_I |D(\tau \circ h_0 - \tau \circ h_1)| \geq d_0 \lambda.$$

If

$$\frac{2M\lambda^2}{1-\lambda} \leq \frac{1}{2}d_0\lambda,$$

then $\inf_I |\Psi'_\infty| \geq d_0\lambda/2$.

Proof. The derivative of the sum of all terms with $k \geq 2$ is bounded by $2M \sum_{k \geq 2} \lambda^k = 2M\lambda^2/(1 - \lambda)$. Subtract this from the first-term lower bound. $\square$

This changes the next billiard calculation from an infinite statistical problem into a finite geometric one. Choose two central return words whose one-step radial tangency germs have different curvature coefficients in (24); attach fixed regular connectors to the magnet; and compute Ψ'_∞ (or a rigorously bounded finite approximation plus its geometric tail). A nonzero lower bound is open under small table perturbations and gives the uniform UNI input for Lemma 6.5. The non-equality of the two one-step curvatures is a promising seed, but it is not by itself the required lower bound because the stable coboundary and all subsequent return terms enter Ψ_∞ . After accelerating both branches, Corollary 6.7 shows that it is enough to retain a lower bound of order λ for the first-term curvature separation; the remaining infinite temporal-distance tail is only $O(\lambda^2)$. This remains a useful branchwise fallback. The next calculation gives a more direct fixed periodic witness for the smooth finite truncations.

6.6 An exact Anosov-cocycle witness from a normal two-cycle

There is a more direct way to certify the geometric nonlinearity needed by the smooth finite truncations. Leclerc's criterion for a smooth area-preserving Axiom-A surface map says that QNL follows if the stable or unstable distribution is not C^2 . At a hyperbolic fixed point, this can be checked by the Anosov cocycle. In determinant-one adapted coordinates, write the map as (F, G) with derivative $\text{diag}(\lambda, \mu)$, where $\lambda > 1$, $\mu = \lambda^{-1}$. The relevant scalar is

$$\mathfrak{A}(F, G) = \frac{G_{xyy}}{\mu} - \frac{F_{xy}G_{xy}}{\lambda - 1} - \frac{G_{xx}F_{yy}}{1 - \mu^3} - \frac{G_{xy}G_{yy}}{\mu^2}, \quad (28)$$

with all derivatives evaluated at the fixed point.

Proposition 6.8 (Exact circular normal-cycle witness). *Consider two unit circular scatterers whose facing boundary points are separated by a unit free-flight gap. Let $\mathcal{R} = T^2$ be the billiard return map to the first component near the unobstructed normal period-two orbit, in Birkhoff coordinates (s, p) with $p = \sin \varphi$. Then*

$$D\mathcal{R}(0) = \begin{pmatrix} 7 & 4 \\ 12 & 7 \end{pmatrix}.$$

For the determinant-one eigenbasis

$$S = \begin{pmatrix} 1 & -\sqrt{3}/6 \\ \sqrt{3} & 1/2 \end{pmatrix}, \quad \det S = 1,$$

write $S^{-1}\mathcal{R}S = (F, G)$. Then

$$\begin{aligned} \lambda &= 7 + 4\sqrt{3}, & \mu &= 7 - 4\sqrt{3}, \\ F_{xy} &= F_{yy} = G_{xx} = G_{xy} = G_{yy} = 0, & G_{xyy} &= -\mu. \end{aligned}$$

Consequently

$$\mathfrak{A}(F, G) = -1.$$

In particular, the local return dynamics has a nonzero QNL witness which is stable under sufficiently small clean perturbations of the table.

Proof. Place the circle centres at $(0, 0)$ and $(3, 0)$ and parametrize the first boundary near the facing point by $q(s) = (\cos s, \sin s)$. For outgoing Birkhoff data (s, p) , set

$$n = (\cos s, \sin s), \quad t = (-\sin s, \cos s), \quad v = \sqrt{1 - p^2} n + pt.$$

For a ray based at q and a target circle with centre c and radius one, put $w = q - c$, $b = w \cdot v$, and $c_0 = |w|^2 - 1$. The first collision time and reflected velocity are

$$t_* = -b - \sqrt{b^2 - c_0}, \quad v^+ = v - 2(v \cdot n_*)n_*, \quad n_* = q + t_*v - c.$$

Apply this formula first to the second circle and then to the first. Taylor expansion through degree three gives the displayed linear matrix. Reflection in the line of centres conjugates (s, p) to $(-s, -p)$ and commutes with the two-collision return, so the return map is odd and all its quadratic jets vanish. Conjugating the cubic jet by S gives $G_{xyy} = -(7 - 4\sqrt{3})$; substitution in (28) gives -1 . $\square$

The calculation is scale free: if both radii and the free-flight gap are multiplied by the same positive number, use dimensionless arclength s/R and obtain the same return jet. Thus the local configuration can be inserted at an arbitrarily small geometric scale into a finite-horizon table, provided the normal two-cycle remains unobstructed. More generally, on any fixed clean finite itinerary the return jet and $\mathfrak{A}$ depend real-analytically on the circular radii, centres, and collision points. Proposition 6.8 therefore implies that the nonzero-cocycle tables form an open dense subset of every connected analytic parameter chart containing this sample.

For an anchored finite horseshoe which contains this normal two-cycle and a regular connector branch, the same fixed point occurs in every larger anchored truncation. Hence the geometric nonlinearity witness itself no longer deteriorates with truncation rank. What still requires proof is a uniform smooth extension/Markov realization of those finite billiard cores, uniform cutting mass for the SRB conditionals, and transfer of Leclerc's QNL bound from the equilibrium state to the weighted double-centered face slice. The exact cocycle closes the local geometry, not these singular-exhaustion steps.

6.7 Transfer of QNL to a central weighted face slice

The equilibrium-state/face-current mismatch is harmless on one compact homogeneous return rectangle. Let W^u be a reference unstable leaf and let $H : \Gamma \rightarrow W^u$ be stable holonomy from a central smooth face arc. Write m_Γ for arclength and let w include the physical face velocity, coarea factor, and the smooth branch Jacobians.

Lemma 6.9 (Weighted face-slice transfer). *Assume, uniformly on a compact clean family, that*

- (i) H is a $C^{1+\alpha}$ diffeomorphism onto its image and $0 < c_H \leq JH \leq C_H$;
- (ii) the SRB conditional μ^u on W^u has arclength density between $c_u > 0$ and $C_u < \infty$;
- (iii) $w \in C^\alpha(\Gamma)$ and $\|w\|_\infty \leq C_w$.

Define the signed face measure

$$\nu = H_*(w \, dm_\Gamma).$$

If, on every interval $J \subset W^u$, the temporal-distance phase X satisfies

$$\mu^u\{t \in J : |X(t) - P(t)| \leq \epsilon\} \leq C_0 \epsilon^\Gamma \mu^u(J),$$

then

$$|\nu|\{t \in J : |X(t) - P(t)| \leq \epsilon\} \leq C_1 \epsilon^\Gamma |J|, \quad C_1 = C_0 \frac{C_w C_u}{c_H c_u},$$

after increasing C_1 by a uniform chart constant. The same conclusion holds for a finite signed sum of physical face pieces, with C_1 multiplied by the sum of their total-variation bounds.

Proof. The change-of-variables formula gives

$$\frac{d|\nu|}{dm_{W^u}}(Hz) = \frac{|w(z)|}{JH(z)} \leq \frac{C_w}{c_H}.$$

The density bounds for μ^u compare arclength and μ^u in both directions. Apply the assumed small-value estimate and make those two comparisons. Total variation and the triangle inequality handle signed pieces. $\square$

Lemma 6.3 supplies the central coarea transversality, while standard clean-rectangle holonomy estimates supply (i)–(ii). Thus, once the exact cocycle is placed in an anchored smooth horseshoe, the central weighted face slice inherits the same QNL exponent. For pieces whose return word or homogeneity rank tends to infinity, this local transfer is combined below with the central marked moment and the two-sided recovery sum; Proposition 7.5 keeps those constants on one common refinement.

6.8 A countable-IFS theorem removes the return truncation

There is a more direct route than passing through smooth finite horseshoes. Leclerc–Paukkonen–Sahlsten [15] prove power Fourier decay for Gibbs measures of finite or countable $C^{1+\alpha}$ conformal IFSs, allowing overlaps. In the countable case their assumptions are uniform contraction, bounded distortion, summable variations, and the light-tail condition

$$\sum_a \sup_{x \in I} e^{\phi(h_a x)} |h'_a(x)|^{-\gamma_0} < \infty \quad \text{for some } \gamma_0 > 0. \quad (29)$$

Their Theorem 1.1 says that QNL gives power Fourier decay, including for Hölder amplitudes and nonstationary phases with a small polynomial loss in their norms. Sections 6.1–6.2 prove QNL by finite-difference cutting arguments for finite MNL and for countable UNI systems with a light tail. No finite-alphabet exhaustion or smooth extension of the branch maps is used.

This theorem is unusually well matched to the present induced problem. Let $F : I \rightarrow I$ be the unstable quotient of a full-branch Gibbs–Markov return to a central billiard magnet and let $\{h_a : I \rightarrow I\}_{a \in \mathcal{A}}$ be its inverse branches. After accelerating once if necessary, the desired dictionary is

countable-IFS input	billiard input	
h_a	homogeneous inverse return branch	
ϕ	normalized induced unstable-Jacobian potential	
uniform contraction	uniform hyperbolicity after inducing	(30)
bounded distortion	homogeneous return distortion	
light tail	return tail plus a small grazing moment	
QNL	a fixed clean finite-difference gate.	

There is a small but important point in the first line of this dictionary. The one-sided moving-face labels must not be inserted by cutting every return branch into labelled

subbranches: those pieces need not have full image and would destroy the hypothesis to which the countable-IFS theorem is applied. The correct object is a full-branch dynamical quotient with a marked amplitude fiber.

For an interval amplitude use the affine scale-normalized norm

$$\|w\|_{\mathcal{A}^\alpha(J)} := \|w\|_{\infty, J} + |J|^\alpha [w]_{C^\alpha(J)}. \quad (31)$$

It is unchanged up to a fixed factor by affine normalization and cannot increase under restriction. Every connected component is charged separately; hence endpoint and component complexity is visible in the sum rather than hidden in a piecewise-Hölder notation.

Proposition 6.10 (Marked full-branch quotient). *Let Λ be a Young return rectangle with product structure and let $\{\Lambda_a\}_{a \in \mathcal{A}}$ be a return partition such that $F|_{\Lambda_a} = T^{R_a}$ maps the stable subset Λ_a across Λ with the usual stable/unstable Markov property. Fix a reference unstable leaf I and let $\pi^s : \Lambda \rightarrow I$ be stable holonomy. Then*

$$\bar{F} \circ \pi^s = \pi^s \circ F$$

defines a one-state full-branch expanding map: for $I_a = \pi^s(\Lambda_a)$, the map $\bar{F} : I_a \rightarrow I$ is onto and has an inverse branch $h_a : I \rightarrow I_a$.

Suppose now that the lift of a complete physical face through Λ_a is refined into one-sided marked pieces $E_{a,\ell}$, after artificial-face cancellation. On the stable quotient its action can be written

$$\nu(\varphi) = \sum_{a,\ell} \int_{J_{a,\ell}} \varphi(h_a x) w_{a,\ell}(x) d\mu^u(x), \quad (32)$$

where the $J_{a,\ell} \subset I$ are amplitude pieces and the sign is included in $w_{a,\ell}$. Thus the LPS dynamical alphabet remains $\{h_a\}_{a \in \mathcal{A}}$; the face label ℓ is a fiber mark and does not have to define a full-image branch. If

$$\sum_{a,\ell} \|w_{a,\ell}\|_{\mathcal{A}^\alpha(J_{a,\ell})} \mu^u(J_{a,\ell}) < \infty,$$

then the marked current is a summable piecewise-Hölder amplitude over the same full-branch Gibbs system.

Proof. The Markov property says that every unstable leaf in Λ_a is mapped across every stable leaf of Λ . Quotienting stable leaves therefore makes $\bar{F}|_{I_a}$ onto I ; contraction of inverse branches and bounded distortion are the usual quotient estimates. Disintegrate the coarea face measure first over return atoms and then over its one-sided physical marks. Stable holonomy transports each piece to the reference leaf and gives (32). No restriction $h_a|_{J_{a,\ell}}$ is declared to be a dynamical branch. The last assertion is the triangle inequality for the Hölder-amplitude norm. Matched artificial cuts contribute opposite weights and disappear before this disintegration. $\square$

Consequently the standard Young quotient and the moving-face bookkeeping are compatible provided the marked-amplitude sum is finite. The construction of the quotient is standard. Proposition 6.11 supplies this summability on the central return alphabet, while the bi-recovery refinement supplies the grazing tail.

The SRB potential part of this audit is now supported by a direct billiard result. Lemmas 5.2–5.3 of Lima–Obata–Poletti [17] prove that the collision Lebesgue/SRB measure is an equilibrium state of the geometric potential

$$\phi = -\log \|DT|_{E^u}\|, \quad P(\phi) = 0,$$

and that ϕ becomes Hölder on their countable Markov coding for finite-horizon dispersing billiards. Hence the induced potential has summable variations. This does not by itself give the one-state return tail, but it removes the former ambiguity between an MME coding and the required collision-SRB Gibbs weights.

Proposition 6.11 (Central amplitude moment). *Assume the Young return scheme has*

$$\sum_a \sup_I p_a e^{\epsilon R_a} < \infty$$

and restrict first to a compact regular homogeneity core. For a complete first moving-face defect, suppose every collision insertion has uniformly bounded C^α coarea and stable-holonomy factors. Then the marked amplitude disintegration (32) satisfies

$$\sum_{a,\ell} e^{\epsilon' R_a} \|w_{a,\ell}\|_{\mathcal{A}^\alpha(J_{a,\ell})} \mu^u(J_{a,\ell}) < \infty \quad (0 < \epsilon' < \epsilon).$$

After high-homogeneity insertions are assigned to the bi-recovery blocks of Corollary 4.4, the same conclusion holds for the entire assembled first defect.

Proof. A return word of length R_a has at most R_a possible physical insertion times. On the compact regular core each insertion contributes at most a uniform constant times the Gibbs branch weight. Hence the left-hand side is bounded by

$$C \sum_a R_a e^{\epsilon' R_a} \sup_I p_a,$$

which is finite because $Re^{\epsilon' R} \leq C_{\epsilon,\epsilon'} e^{\epsilon R}$. Artificial insertions cancel before taking total variation. The removed high-homogeneity pieces satisfy the common two-sided recovery moment and are summed by Proposition 4.1. $\square$

The following consequence records the exact countable-IFS interface rather than hiding it in a finite-horseshoe approximation.

Proposition 6.12 (Finite-difference countable-IFS bridge). *Let the central unstable quotient be a one-state full-branch $C^{1+\alpha}$ IFS with uniform inverse contraction, bounded distortion, a normalized Gibbs potential of summable variations, and (29). For words $\mathbf{a}, \mathbf{b}$ of length N and $x, y \in I$, put*

$$\Delta_N((\mathbf{a}, x), (\mathbf{b}, y)) = \{\log |h'_\mathbf{a}(x)| - \log |h'_\mathbf{b}(x)|\} - \{\log |h'_\mathbf{a}(y)| - \log |h'_\mathbf{b}(y)|\}.$$

Assume the LPS quantitative nonlinearity estimate: for some $C > 0$, $\Theta > 0$, and $\rho \in (0, 1)$,

$$\sum_{\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d} \in \mathcal{A}^N} p_\mathbf{a} p_\mathbf{b} p_\mathbf{c} p_\mathbf{d} \mathbf{1}_J(\Delta_N((\mathbf{a}, x_\mathbf{c}), (\mathbf{b}, x_\mathbf{d}))) \leq C\{|J|^\Theta + \rho^N\} \quad (33)$$

for every interval $J \subset \mathbb{R}$; Gibbs weights may be evaluated at arbitrary reference points in their cylinders. Then the induced SRB Gibbs measure has the weighted nonstationary-phase Fourier estimate of Leclerc–Paukkonen–Sahlsten for every phase with the stated C^α log-derivative bounds. The conclusion remains valid for the marked physical face amplitudes of Lemma 6.9.

Proof. Equation (33) is precisely the QNL hypothesis in Theorem 1.1 of [15]. That theorem is formulated for finite or countable $C^{1+\alpha}$ systems and Hölder amplitudes; it neither differentiates $\log h'_a$ nor assumes a C^2 holonomy. The light-tail moment controls the countable alphabet. The weighted-face conclusion follows from Lemma 6.9 and the summable scale-normalized amplitude norm (31). $\square$

The former C^2 multiplier/UNI route is deliberately not used. For a billiard quotient the stable holonomy is generally only $C^{1+\alpha}$, so the derivative of $\log h'_a$ need not exist. The correct gate input is the finite-difference estimate (33).

Proposition 6.13 (Clean homoclinic horseshoe supplies a finite QNL gate). *Let p be the regular fixed point of the squared billiard map associated with the normal period-two orbit in Proposition 6.8. If its Anosov cocycle satisfies $\mathfrak{A}(p) \neq 0$, then there is a compact finite clean horseshoe Ω_p , contained in regular branches and containing p , such that in at least one time orientation its temporal distance satisfies QNL for every Hölder equilibrium state. Thus Ω_p supplies a finite topological QNL seed; Proposition 6.19 transfers its polynomial escape to the physical collision-SRB bracket.*

Proof. Lemma 5.1 of Lima–Obata–Poletti [17] says that finite-horizon dispersing billiards have a single homoclinic class at regular hyperbolic points. Choose a second regular hyperbolic periodic point q . The theorem gives clean transverse intersections $W^u(p) \pitchfork W^s(q)$ and $W^u(q) \pitchfork W^s(p)$. The inclination lemma then gives a nontrivial clean transverse homoclinic point of p ; this avoids the vacuous substitution $x = y = p$. The Smale–Birkhoff construction, using a finite segment of this clean homoclinic orbit, gives a compact finite horseshoe Ω_p on which the billiard map is smooth. In symplectic Birkhoff coordinates the map is area preserving; the local dynamics near Ω_p may be viewed as a smooth Axiom–A basic set.

By the contrapositive of the Anosov cocycle obstruction, together with Theorem 4.3 of Leclerc [14], in one of the two time orientations the temporal distance on Ω_p satisfies QNL for every Hölder equilibrium state, in particular for the equilibrium state of the geometric potential restricted to this finite subsystem. No comparison of that horseshoe measure with collision SRB is made here. $\square$

There is a regularity caveat. The stable quotient of a smooth horseshoe is in general only $C^{1+\alpha}$, precisely because the stable holonomy need not be C^2 . Therefore the finite QNL conclusion above cannot be fed directly into the C^2 non-linearizability criterion of Algom–Rodriguez Hertz–Wang. The route below instead propagates the physical gate’s finite-difference gap to the countable $C^{1+\alpha}$ inducing scheme and then invokes LPS in its native regularity.

The light-tail hypothesis has a concrete billiard reduction. If a branch a has return time R_a and grazing rank k_a , bounded distortion gives, schematically,

$$|h'_a|^{-\gamma_0} \leq C e^{c\gamma_0 R_a} (1 + k_a)^{q\gamma_0}.$$

Thus an exponential return moment and any strictly positive polynomial SRB tail in k_a imply (29) after choosing $\gamma_0 > 0$ sufficiently small. This is strictly weaker than uniform control of every rank- L smooth extension.

Lemma 6.14 (Return and grazing moments imply the LPS light tail). *Assume the normalized branch weights $p_a(x) = e^{\phi(h_a x)}$ satisfy*

$$\sum_a \sup_x p_a(x) e^{\epsilon R_a} (1 + k_a)^\delta < \infty$$

for some $\epsilon, \delta > 0$, and suppose $|h'_a|^{-1} \leq C e^{cR_a} (1 + k_a)^q$. Then (29) holds for every

$$0 < \gamma_0 < \min\{\epsilon/c, \delta/q\}.$$

Proof. Raise the inverse-derivative estimate to γ_0 and absorb the two factors into the assumed exponential and polynomial moment. $\square$

There is also a pressure-based way to obtain the required moment. The geometric-potential theory for finite-horizon dispersing billiards has a spectral gap for $-t \log J^u T$ on every compact subinterval of $(0, t_*)$, where $t_* > 1$, and its modified one-step expansion is uniform for $t \geq t_0 > 0$ [2]. Since the SRB value is $t = 1$, the LPS factor $|h'_a|^{-\gamma_0}$ merely changes the geometric exponent from 1 to $1 - \gamma_0$.

Proposition 6.15 (Geometric-pressure route to the light tail). *Let a homogeneous full-branch inducing scheme have an exponential return tail at the SRB parameter. Assume its induced distortion is uniform and that the geometric transfer estimates hold for $t \in [1 - \varepsilon, 1]$. Then, after decreasing $\varepsilon > 0$ if necessary, its normalized induced SRB potential satisfies (29) for every sufficiently small $\gamma_0 > 0$.*

Proof. Group branches by return time $R_a = n$. Up to the uniform distortion coboundary, the summand in (29) is the inverse unstable Jacobian to the power $1 - \gamma_0$. The geometric transfer estimate at $t = 1 - p\gamma_0$ bounds the corresponding p -moment by $Ce^{c(p\gamma_0)n}$, with $c(p\gamma_0) \rightarrow 0$ as $\gamma_0 \rightarrow 0$. Hölder's inequality combines this with the SRB return tail Ce^{-an} to give

$$\sum_{R_a=n} \sup_x e^{\phi(h_a x)} |h'_a(x)|^{-\gamma_0} \leq Ce^{-an/q + c(p\gamma_0)n/p}.$$

Choose $p, q > 1$ conjugate and then γ_0 so that the exponent is negative; summing over n proves the claim. Homogeneity-strip subdivision is already included in the geometric one-step expansion. $\square$

Lemma 6.16 (Fixed-gate hitting and two-context moments). *Let F be a one-state full-branch Gibbs–Markov map and let $\mathcal{G}$ be a fixed finite gate cylinder of return depth L . Suppose bounded distortion gives*

$$\inf_C \mu^u(\mathcal{G} \mid C) \geq \eta_G > 0 \tag{34}$$

for every past cylinder C , and suppose (82) holds for some $\epsilon > 0$. If N_G is the first gate-hitting time in L -blocks and C^+ is the total collision length of the future entrance context, then there is $\epsilon_G > 0$ such that

$$\mu^u(N_G > k) \leq (1 - \eta_G)^k, \quad \mathbb{E}e^{\epsilon_G C^+} < \infty.$$

The reverse quotient gives the same estimate for C^- . On the natural extension no independence is needed:

$$\mathbb{E}e^{\epsilon'_G(C^- + C^+)} < \infty \tag{35}$$

for every sufficiently small $\epsilon'_G > 0$.

Proof. Condition (34) and the Markov property give $\mu^u(N_G > k + 1 \mid N_G > k) \leq 1 - \eta_G$, proving the geometric tail. Let $K(\theta)$ be the uniform conditional exponential moment of one return; $K(0) = 1$ and $K(\theta) < \infty$ for small $\theta > 0$. Bounded Gibbs distortion gives

$$\mathbb{E}\left[e^{\theta C^+}; N_G = k\right] \leq C\{(1 - \eta_G)K(\theta)^L\}^{k-1}K(\theta)^L.$$

Choose θ so that the quantity in braces is smaller than one and sum in k . Time reversal proves the past estimate. Finally

$$\mathbb{P}(C^- + C^+ > t) \leq \mathbb{P}(C^- > t/2) + \mathbb{P}(C^+ > t/2),$$

so the two marginal exponential tails imply (35) without independence. $\square$

For the marked quotient, choose one fixed finite gate word. The Gibbs ratio of that word after an arbitrary prefix differs from its unconditional weight by at most the uniform distortion factor. Hence (34) holds with $\eta_G = C_D^{-1}\mu^u(\mathcal{G}) > 0$; it is not an additional mixing assumption.

Lemma 6.17 (Joint return–maximal-grazing moment). *Let Y be the base of an exponential Young return, R its return time, and let $B(x)$ be the largest homogeneity rank visited before $R(x)$. In $(r, p) = (r, \sin \varphi)$ coordinates suppose*

$$\mu\{1 - |p| \lesssim 2^{-2b}\} \leq C2^{-2b}, \quad \mu_Y(R > L) \leq Ce^{-cL}.$$

Then B has an exponential tail. More precisely, for every fixed $A > 0$,

$$\mu_Y(B \geq b) \leq Ce^{-cAb} + CA b 2^{-2b}. \quad (36)$$

Consequently, for sufficiently small $\epsilon, \kappa > 0$,

$$\mathbb{E}_Y e^{\epsilon R + \kappa B} < \infty. \quad (37)$$

If the inverse-Jacobian and coarea loss of a return word is bounded by $Ce^{c_0 R + qB}$, then (29) and the marked-amplitude moment hold after decreasing the exponents. No independence of R and B is required.

Proof. On $\{R \leq Ab\}$ take a union bound over the at most Ab collision times. Since $\mu_Y(E) \leq \mu(E)/\mu(Y)$ for every event E and the collision SRB measure is invariant, each visit to a rank- b boundary layer costs at most $C2^{-2b}$. Adding $\mu_Y(R > Ab)$ proves (36). Choose A so that both terms decay exponentially in b ; hence $\mathbb{E}e^{q\kappa B} < \infty$ for some $q > 1$ and small κ . For conjugate $p, q > 1$, Hölder gives

$$\mathbb{E}e^{\epsilon R + \kappa B} \leq (\mathbb{E}e^{p\epsilon R})^{1/p} (\mathbb{E}e^{q\kappa B})^{1/q} < \infty.$$

Absorb $c_0 R + qB$ with smaller exponents and apply Lemma 6.14. The same argument applies to the scale-normalized coarea amplitude. $\square$

Lemma 6.17 verifies the grazing factor from the collision density and strip width, while Proposition 6.11 supplies the moving-face amplitude budget on the same quotient. Lemma 6.16 supplies the entrance/exit moment. Thus the particular full-branch quotient uses one joint return/grazing/context ledger rather than an unverified appeal to the literature theorem.

This route also exposes a flaw in the finite-horseshoe detour. A conservative pasting lemma [19] can locally modify or smooth a volume-preserving map, but it does not automatically extend a finite family of billiard branches to a global area-preserving Axiom–A diffeomorphism whose equilibrium state has the required collision-SRB weights. Even a successful extension can put a fractal Gibbs conditional on a saddle basic set in place of the absolutely continuous unstable conditional of the billiard SRB measure. The direct countable-IFS quotient preserves the correct measure and should therefore be the primary route.

Thus light-tail is not expected to be a new principal theorem once the standard homogeneous inducing scheme is put in the one-state form required above. The exact value $\mathfrak{A} = -1$, the clean homoclinic theorem, and Proposition 6.13 produce the clean polynomial escape seed without naming a connector. Proposition 6.23 records the local physical slope transfer, and the cohomological bracket below identifies its physical collision-SRB object.

6.9 The cohomological bracket and polynomial context blow-ups

The quotient regularity is only $C^{1+\alpha}$. The useful exact object is therefore a four-point difference of Birkhoff sums, not a derivative of a holonomy. Let f denote the billiard map restricted to the finite clean basic set, let F be its stable factor, and put $\tau_f = \log J^u f$ and $\tau_F = \log |F'|$. As in [14, Lemma 3.15], choose a Hölder function θ such that

$$\tau_f = \tau_F \circ \pi + \theta \circ f - \theta. \quad (38)$$

For p, q in one Markov rectangle define

$$\Delta(p, q) = \sum_{j \in \mathbb{Z}} \{\tau_f(f^j p) - \tau_f(f^j[p, q]) - \tau_f(f^j[q, p]) + \tau_f(f^j q)\}. \quad (39)$$

Lemma 6.18 (Cohomological bracket identification). *Let $\mathbf{a}, \mathbf{b}$ be the length- N past words of p, q and let $x = \pi p$, $y = \pi q$. With inverse branches $g_{\mathbf{a}}, g_{\mathbf{b}}$ of F , set*

$$\begin{aligned} \mathcal{D}_N(\mathbf{a}, \mathbf{b}; x, y) &:= S_N \tau_F(g_{\mathbf{a}} x) - S_N \tau_F(g_{\mathbf{a}} y) \\ &\quad - S_N \tau_F(g_{\mathbf{b}} x) + S_N \tau_F(g_{\mathbf{b}} y). \end{aligned} \quad (40)$$

Then

$$|\mathcal{D}_N(\mathbf{a}, \mathbf{b}; x, y) - \Delta(p, q)| \leq C \kappa^{\alpha N}. \quad (41)$$

The constants are uniform on the persistent finite basic set and for $|s| < s_0$. Entrance and exit contexts are not included in this statement; their finite jets and holonomy tails are treated explicitly in Proposition 6.19.

Proof. Insert (38) in (39). The four θ -terms telescope. For $j \geq 0$, stable projection gives $\pi(f^j p) = \pi(f^j[p, q])$ and $\pi(f^j q) = \pi(f^j[q, p])$, so the positive-time factor terms cancel exactly. Writing the negative iterates through the past inverse words gives (40). The omitted tail starts at $N + 1$; Hölder regularity of τ_F and inverse contraction give

$$\sum_{j > N} C \kappa^{\alpha j} \leq C' \kappa^{\alpha N}.$$

This is precisely the proof of [14, Lemma 3.15] on the clean basic set. Hyperbolic continuation makes the constants uniform in s . $\square$

The next statement transfers the all-scale escape in [14, Propositions 4.31–4.32] from the horseshoe conditional to the physical collision-SRB conditional. This is the only support transfer that is required.

The preceding bracket must still be matched to the physical endpoint maps. For a stable projection π_s on a clean rectangle, the stable holonomy Jacobian satisfies the convergent identity

$$\log J\pi_s(r) = \sum_{j \geq 0} \{\tau_f(f^j r) - \tau_f(f^j \pi_s r)\}. \quad (42)$$

This is the Jacobian formula of [14, Lemma C.1]. In particular, entrance and exit holonomies are not discarded as unspecified coboundaries: their four endpoint series are inserted into (39); the common terms cancel and the unpaired tail is bounded by $C \kappa^{\alpha N}$.

Proposition 6.19 (Context-polynomial blow-up). *Assume $\mathfrak{A}(p) = -1$ and let $I_c = t_c + \ell_c[-1, 1]$ be any homogeneous resolving cylinder whose central excursion enters the persistent clean basic set. For the full reduced-slope bracket S_c , including the two endpoint*

holonomy Jacobians, let N be the total central resolving depth and let $L_c = |c^-| + |c^+|$ be the entrance/exit length. There are a fixed degree d , a rough template $\mathcal{X}_{\omega_c}$ from the clean basic set, a polynomial P_c of degree at most d , and a context cost $H_c \geq 1$ such that

$$\ell_c^{-1}\{S_c(t_c + \ell_c u) - S_c(t_c)\} = \mathcal{X}_{\omega_c}(u) + P_c(u) + E_c(u), \quad \|E_c\|_\infty \leq C \left(H_c \ell_c^\alpha + \frac{\kappa^{\alpha N}}{\ell_c} \right). \quad (43)$$

Moreover $H_c \leq C e^{aL_c}$, and for some $p > 1$ the context ledger gives

$$\sup_{|s| < s_0} \sum_c w_c H_c^p < \infty. \quad (44)$$

The degree and the escape constants of $\mathcal{X}_{\omega_c}$ modulo $\mathbb{R}_d[X]$ are independent of c and s . In particular, if

$$N \geq q_* |\log \ell_c|, \quad q_* > \frac{1 + \alpha}{\alpha |\log \kappa|}, \quad (45)$$

then the second error in (43) is $O(\ell_c^\alpha)$.

Proof. Use (42) for both endpoint maps and split each series at the entrance and exit times. The four common tails cancel in the bracket. Each remaining finite clean context sum is $C^{1+\alpha}$ on its homogeneous component. Its affine Taylor polynomial is linear (and is regarded as an element of $\mathbb{R}_d[X]$); the normalized remainder is bounded by its $C^{1+\alpha}$ norm times ℓ_c^α . The chain rule and uniform one-step distortion give $H_c \leq C e^{aL_c}$. Choose $p > 1$ so that pa is smaller than the exponent in (35); this proves (44). Lemma 6.18 identifies the central part with the factor four-point cocycle with error $O(\kappa^{\alpha N})$. Condition (45), rather than an unrecorded fixed acceleration, makes its normalized value $O(\ell_c^\alpha)$. Finally [14, Propositions 4.31–4.32], activated by $\mathfrak{A}(p) \neq 0$, gives uniform all-location, all-scale escape of the rough template from every polynomial of degree at most d . No fixed acceleration is used. $\square$

6.10 Gate-resolved conditional slope energy

The bracket furnished by Lemma 6.18 is intrinsically a four-point object. We therefore use it as a conditional self-difference and recover a one-point small-ball estimate only at the end by Jensen. This avoids both an unjustified one-branch cancellation and the unnecessary countable four-tree QNL theorem. Write α_B for the billiard remainder exponent in the physical $C^{1+\alpha_B}$ charts.

Proposition 6.20 (Exact endpoint bracket on a physical cell). *Let $\kappa : U_\beta \rightarrow I$ be the physical face coordinate and let $\mathbf{a}_\beta^-, \mathbf{a}_\beta^+$ be the normalized past and future inverse words. With endpoint holonomies $A_\pm$, put*

$$X = A_- \circ h_- \circ \kappa, \quad Y = A_+ \circ h_+ \circ \kappa, \quad S_\beta = \log |Y'/X'|.$$

For $t, t' \in U_\beta$, set $x_t = \kappa(t)$. Then

$$S_\beta(t) - S_\beta(t') = \mathcal{D}_N(\mathbf{a}_\beta^+, \mathbf{a}_\beta^-; x_t, x_{t'}) + C_\beta(t) - C_\beta(t') + \mathcal{R}_{\beta, N}(t, t'), \quad (46)$$

$$\|\mathcal{R}_{\beta, N}\|_\infty \leq C \kappa_+^{\alpha_B N}. \quad (47)$$

Here $C_\beta = c_\beta^+ - c_\beta^-$ is the sum of the finite entrance and exit contexts and $\|C_\beta\|_{C^{1+\alpha_B}} \leq H_\beta$. The signs in the bracket are, in order,

$$(+, \mathbf{a}_\beta^+, t), \quad (-, \mathbf{a}_\beta^+, t'), \quad (-, \mathbf{a}_\beta^-, t), \quad (+, \mathbf{a}_\beta^-, t').$$

Let $Q = t_Q + \ell_Q[-1, 1] \subset U_\beta$. After affine normalization in the t -variable, with t' fixed, there are a clean rough template $\mathcal{X}_{\beta,Q}$, a polynomial $P_{\beta,Q,t'}$ of degree at most d , and an error satisfying

$$S_\beta(t_Q + \ell_Q u) - S_\beta(t') = \ell_Q \{\mathcal{X}_{\beta,Q}(u) + P_{\beta,Q,t'}(u)\} + \mathcal{E}_{\beta,Q,t'}(u), \quad (48)$$

$$\|\mathcal{E}_{\beta,Q,t'}\|_\infty \leq C \{H_\beta \ell_Q^{1+\alpha_B} + \kappa_+^{\alpha_B N}\}. \quad (49)$$

There are physical branch scales $a_X, a_Y > 0$ for which

$$c_X \leq |X'|/a_X, |Y'|/a_Y \leq C_X, \quad [\log(|X'|/a_X)]_{\alpha_B} + [\log(|Y'|/a_Y)]_{\alpha_B} \leq C_X. \quad (50)$$

The constants are uniform in t' , the entrance/exit context, and the small table parameter.

Proof. The chain rule gives

$$S_\beta = \log |h'_+| - \log |h'_-| + \log |A'_+ \circ h_+| - \log |A'_- \circ h_-|.$$

Insert the stable-holonomy Jacobian series at the past endpoint and its time-reversed series at the future endpoint. Subtract the values at t and t' before truncation. The four resulting signs are exactly the displayed sign table. The common infinite endpoint tails cancel; the remaining central terms are

$$[S_{N\tau_F}(g_{\mathbf{a}^+}x_t) - S_{N\tau_F}(g_{\mathbf{a}^+}x_{t'})] - [S_{N\tau_F}(g_{\mathbf{a}^-}x_t) - S_{N\tau_F}(g_{\mathbf{a}^-}x_{t'})],$$

which is $\mathcal{D}_N$, while the unpaired tail is bounded geometrically. This proves (46).

On Q , Taylor expand only the t -dependent finite context; all t' -terms are constants and enter the constant coefficient of $P_{\beta,Q,t'}$. Proposition 6.19 identifies the t -dependent central part with the clean rough template modulo a degree- d polynomial. The Taylor remainder is $H_\beta \ell_Q^{1+\alpha_B}$, proving the last display. Thus the four-point object, rather than a one-branch cancellation, is used throughout. $\square$

Definition 6.21 (Physical stopping cover and marked conditional). *Let μ_β be the normalized collision-SRB conditional on U_β , and let $\nu_\beta = (S_\beta)_* \mu_\beta$. For a scale ℓ , $\mathcal{P}_{\beta,\ell}$ is the deterministic physical interval partition into consecutive cells of length ℓ , with the last cell merged into its predecessor. Legal clean cylinders are constructed inside each physical parent. Hence every $Q \in \mathcal{P}_{\beta,\ell}$ satisfies*

$$c_0 \ell \leq |Q| \leq C_0 \ell, \quad \sum_{Q \in \mathcal{P}_{\beta,\ell}} \mu_\beta(Q) = 1,$$

with bounded overlap after the harmless endpoint enlargement. A marked conditional has the form $d\eta_\beta = a_\beta d\mu_\beta$, where $\|a_\beta\|_\infty + |U_\beta|^\alpha [a_\beta]_\alpha \leq \text{Mark}(\beta)$.

Lemma 6.22 (Gate, buffer, and total-time allocation). *Let N be the total available induced depth. Fix $0 < \alpha_L < \alpha_B$ and choose $h, s > 0$ with*

$$\frac{1+h}{1+\alpha_B} < s < \frac{1}{1+\alpha_L}. \quad (51)$$

Let c_* be the deletion-depth constant in Proposition 6.23. Choose $b_0, q, \zeta > 0$ so that

$$qb_0 < \zeta < 1, \quad q\alpha_B |\log \kappa_+| > 1, \quad c_* s b_0 < 1 - \zeta. \quad (52)$$

For $r_N = \max\{r, e^{-b_0 N}\}$, put

$$\ell = r_N^s, \quad m = \lceil q |\log r_N| \rceil, \quad n = N - m.$$

Then $m \leq \zeta N + 1$. All length- m continuations are retained, and on contexts $H_\beta \leq r_N^{-h}$,

$$H_\beta \ell^{1+\alpha_B} + \kappa_+^{\alpha_B m} \leq C r_N, \quad r_N \leq \ell^{1+\alpha_L}. \quad (53)$$

Moreover

$$\sum_{\tau_G > n} w_\beta + \sum_{H_\beta > r_N^{-h}} w_\beta \leq C e^{-cn} + C r_N^{hp}. \quad (54)$$

If later $r = \Xi^{-\delta}$ and $\log \Xi \leq \chi_+ N$, choose additionally $q\delta\chi_+ < \zeta$ and $c_* s\delta\chi_+ < 1 - \zeta$; the buffer and all clean deletion generations then remain inside the same total time N .

Proof. Successive conditioning on the persistent gate in blocks of fixed length gives the first exponential tail. Markov's inequality and (44) give the second. The two inequalities in (51) give respectively $r_N^{-h} \ell^{1+\alpha_B} \leq r_N$ and $r_N \leq \ell^{1+\alpha_L}$. The choice of q controls the holonomy tail. The last inequality reserves $K(\ell) = c_* |\log \ell| + O(1)$ clean generations inside the core n . Since all continuations are retained, their conditional weights sum to one; the buffer changes depth but not mass. $\square$

Proposition 6.23 (Legal-child physical deletion). *There are $\eta, \kappa_g, \gamma_{\text{gate}} > 0$ such that the following holds uniformly. For $Q \in \mathcal{P}_{\beta, \ell}$, fixed $t' \in U_\beta$, and every degree- d polynomial $P_{\beta, Q, t'}$, the legal children in the physical extension of Leclerc's cutting construction satisfy*

$$c_1 \kappa_g |Q| \leq |Q'| \leq C_1 \kappa_g |Q|, \quad \mu_\beta(Q' \mid Q) \geq \eta \quad (55)$$

for the union of escaping children. If $L_k(Q, t')$ is the conditional mass not yet deleted after k generations, then

$$L_{k+1}(Q, t') \leq (1 - \eta) L_k(Q, t') + C \kappa_g^{\gamma_0 k} \mu_\beta(Q). \quad (56)$$

For $K(\ell) = \lceil c_* |\log \ell| \rceil$,

$$\mu_\beta\{t \in Q : |\mathcal{X}_{\beta, Q}(u) + P_{\beta, Q, t'}(u)| \leq C \ell^{\alpha_L}\} \leq C \ell^{\gamma_{\text{gate}}} \mu_\beta(Q). \quad (57)$$

The same estimate with $d\eta_\beta = a_\beta d\mu_\beta$ gains only $C \text{Mark}(\beta)$.

Proof. Leclerc's finite child lemma is applied on the persistent finite clean rectangle. Each selected symbolic child is pulled back through the fixed smooth gate extension. The collision-SRB conditional density and the gate Jacobian are bounded above and below on the homogeneous chart, so symbolic diameter ratios and Gibbs child weights become (55). No Cantor limiting measure is substituted for the physical conditional: only finite cylinders are used.

At generation k , delete the escaping legal children. Children meeting the two endpoints or a gap are stopped and charged to the first ancestor whose diameter is comparable with that gap. Bounded overlap and geometric diameter loss give the error in (56). Iteration yields

$$L_K \leq (1 - \eta)^K \mu_\beta(Q) + C \sum_{j < K} (1 - \eta)^{K-1-j} \kappa_g^{\gamma_0 j} \mu_\beta(Q).$$

Taking $K = c_* |\log \ell| + O(1)$ proves (57). The physical partition in Definition 6.21 has no short terminal cell and overlap one. Multiplication by a_β is controlled by its scaled supremum and Hölder norm. $\square$

Theorem 6.24 (Physical energy tree). *Under the context moment and Lemma 6.22, define*

$$E_\beta(r) = \iint \mathbf{1}_{\{|S_\beta(t) - S_\beta(t')| \leq r\}} d\mu_\beta(t) d\mu_\beta(t'). \quad (58)$$

Then, for $\Theta = \min\{s\gamma_{\text{gate}}, hp\} > 0$,

$$\sum_\beta w_\beta E_\beta(r) \leq Cr^\Theta + Ce^{-cN}. \quad (59)$$

The estimate is local for every marked conditional of Definition 6.21. Consequently, for $\nu_N = \sum_\beta w_\beta \nu_\beta$,

$$\sup_z \nu_N([z - r, z + r]) \leq C\{r^{\Theta/2} + e^{-cN}\}. \quad (60)$$

Proof. Fix t' and partition only the t -variable by $\mathcal{P}_{\beta,\ell}$. Thus pairs with t, t' in different cells are not discarded: the entire t' -dependence is in $P_{\beta,Q,t'}$, uniformly for every Q . On good contexts, (49) and (53) are $O(r_N)$; after division by ℓ , the target radius is at most $C\ell^{\alpha_L}$. Apply (57) on every Q , sum $\sum_Q \mu_\beta(Q) = 1$, and then integrate t' . This gives $C\ell^{\gamma_{\text{gate}}}$. Add the no-gate and bad-context masses from (54), then use $r_N^\Theta \leq r^\Theta + e^{-b_0\Theta N}$. The marked version follows from the last assertion of Proposition 6.23.

For an interval I of radius r ,

$$\nu_\beta(I)^2 \leq (\nu_\beta \otimes \nu_\beta)\{|x - y| \leq 2r\} = E_\beta(2r).$$

Jensen in w_β , followed by (59), gives (60). The total-time allocation in Lemma 6.22 ensures that the added resolution has not changed the two observation times. $\square$

Corollary 6.25 (Central slope small-ball). *The complete central collision-SRB quotient satisfies (60) with constants uniform in s .*

Proof. Apply Theorem 6.24 and the uniform marked return/grazing/context moments. $\square$

Lemma 6.26 (Slope small-ball controls the balanced phase). *For $\psi_\xi = \xi_1 X + \xi_2 Y$ on an affine-normalized gate chart, let*

$$\Xi = |\xi_1|a_X \asymp |\xi_2|a_Y, \quad (61)$$

with a_X, a_Y as in (50). The mass of $\{|\psi'_\xi| \leq \Xi r\}$ is at most $C(r^{\Theta/2} + e^{-cn})$. Smooth value-space cutoffs on its complement have C^α norm $O(r^{-1})$ and satisfy $\inf |\psi'_\xi| \geq c\Xi r$.

Proof. By Proposition 6.20,

$$\psi'_\xi = X'(\xi_1 + \varepsilon_{\text{or}} \xi_2 e^S).$$

The lower and upper bounds in (50) convert the bracketed factor to the physical scale (61). Only opposite-oriented coefficients can resonate, and the resonant set is contained in one interval $|S - z_\xi| \leq Cr$. Apply (60). Composing a fixed smooth three-term partition with $(\xi_1 + \varepsilon_{\text{or}} \xi_2 e^S)/(\Xi r)$ gives the cutoff and norm bounds. $\square$

Lemma 6.27 (Global double projection before disintegration). *Let $\mathcal{G}_\alpha$ be the projective space of signed graph currents with the affine scale-normalized amplitude norm, let $\mathcal{M}_\alpha^\pm$ be the corresponding one-sided marginal spaces, and let $\mathcal{R}_\alpha$ be the regular product-current space. Work in the direct-sum space*

$$\mathfrak{C}_\alpha = \mathcal{G}_\alpha \oplus_\pi (\mathcal{M}_\alpha^- \widehat{\otimes} \nu^+) \oplus_\pi (\nu^- \widehat{\otimes} \mathcal{M}_\alpha^+) \oplus_\pi \mathcal{R}_\alpha, \quad (62)$$

where ν^-, ν^+ are fixed smooth probability references. Let J be the complete signed face current in common physical coordinates. If m_J^-, m_J^+ are its marginals, define $P_- J = m_J^- \otimes \nu^+$ and $P_+ J = \nu^- \otimes m_J^+$. Put

$$J^\circ = (I - P_-)(I - P_+)J = J - m_J^- \otimes \nu^+ - \nu^- \otimes m_J^+ + J(1,1)\nu^- \otimes \nu^+. \quad (63)$$

Then J° is double centered. If the complete response current already has zero marginals, then

$$J^\circ = J. \quad (64)$$

Only after this identity is formed in physical coordinates do we disintegrate the graph component; the two rank-one components and the regular product component remain separate blocks of (62). If the original physical amplitudes obey (103) and the references have fixed C^α densities, then

$$\|J^\circ\|_{\mathfrak{C}_\alpha} \leq C\|J\|_{\mathcal{G}_\alpha}, \quad \sum_\lambda \|J_{\lambda, \text{graph}}^\circ\|_{\mathcal{G}_\alpha} + \|J_{\text{rank}}^\circ\| + \|J_{\text{reg}}^\circ\| < \infty. \quad (65)$$

Proof. The operators commute and $P_- P_+ J = J(1,1)\nu^- \otimes \nu^+$, proving the algebraic formula and centering. Marginalization of one normalized graph piece is bounded from $\mathcal{G}_\alpha$ to $\mathcal{M}_\alpha^\pm$ by Fubini, the scaled C^α norm, and the upper/lower conditional-density bounds. Tensoring with a fixed smooth probability is bounded into the corresponding rank-one summand of (62). It is *not* asserted to produce another graph current. The regular two-dimensional Jacobian/density derivative is placed in $\mathcal{R}_\alpha$. Taking the infimum over projective decompositions and using (103) proves (65). Only the surviving graph summand is refined by face labels; hence no cancellation is postponed until after label-dependent coordinates are chosen. $\square$

Lemma 6.28 (Dominant-coordinate reduction). *After assembly and the recentering of Lemma 6.27, retain the rank-one and regular-product summands in the projective tensor space and evolve both factors by their centered operators. On a clean graph rectangle in the diagonal band $D^{-1}M < N < AM$, the graph-current pairing decomposes as*

$$B_{M,N} = B_{M,N}^{\text{dom,dom}} + B_{M,N}^{\text{dom,rem}} + B_{M,N}^{\text{rem,dom}} + B_{M,N}^{\text{rem,rem}}. \quad (66)$$

The first term is a one-variable integral with physical coordinates X, Y . Every other term is bounded by

$$Ce^{-c \max\{M,N\}} \|g\|_{C^1} \|f\|_{C^1}. \quad (67)$$

The global joint constant and axis blocks vanish or remain in the explicitly separated rank-one summands before the graph-rectangle sum is estimated. Every graph-axis mode created by a boundary cutoff is uniformly nonstationary and is charged to its cutoff norm.

Proof. In stable-unstable endpoint coordinates, freeze the coordinate contracted by the relevant inverse return and subtract the frozen value. The mean-value theorem and hyperbolicity give remainders bounded by $C\lambda_-^M \|g\|_{C^1}$ and $C\lambda_+^N \|f\|_{C^1}$. In the diagonal band these imply (67). The two frozen functions depend only on X, Y , proving (66). Global projection separates the constant and two axis blocks before any label-dependent chart is chosen. They are estimated in their rank-one or regular-product tensor spaces, not redescribed as graph currents; Lemma 6.31 supplies product decay. A graph boundary cutoff can reintroduce an axis term, but it has phase $\xi_1 X$ or $\xi_2 Y$; (50) gives the physical derivative lower bound, and its cutoff ledger is absolutely summable by (65). $\square$

Proposition 6.29 (Scaled physical face density). *On a homogeneous physical face cell $Q = t_Q + \ell_Q[-1, 1]$, the collision-SRB conditional is $d\mu_Q^u = \rho_Q(t) dt$, where, before normalization,*

$$\rho_Q^{\text{raw}}(t) = c_\mu \cos \varphi(t) J_{\text{coa}}(t) J_{\text{hol}}(t) J_{\text{chart}}(t), \quad J_{\text{coa}} = |\nabla G|^{-1}. \quad (68)$$

Put $\widehat{\rho}_Q(u) = \ell_Q \rho_Q(t_Q + \ell_Q u)$. There is a mark

$$\text{Mark}(Q) = 1 + \mathfrak{h}_Q + \mathfrak{b}_Q + [\log J_{\text{hol}}]_{\alpha, Q} + [\log J_{\text{coa}}]_{\alpha, Q} + [\log J_{\text{chart}}]_{\alpha, Q} + \|w_Q\|_{\mathcal{A}^\alpha} \quad (69)$$

such that

$$C^{-1} \text{Mark}(Q)^{-1} \leq \widehat{\rho}_Q \leq C \text{Mark}(Q), \quad [\log \widehat{\rho}_Q]_\alpha \leq C \text{Mark}(Q). \quad (70)$$

For some $p_d > 2$,

$$\sum_Q M_Q \text{Mark}(Q)^{p_d} < \infty \quad (71)$$

uniformly in the small table family. The same statement holds after the positive/negative decomposition of a signed face amplitude.

Proof. In collision coordinates the invariant density is $c_\mu \cos \varphi dr d\varphi$. Coarea along the physical graph, followed by stable projection to the gate coordinate, gives (68). On a homogeneous strip, $\cos \varphi$, the chart Jacobian, and the holonomy Jacobian have the standard scaled log-Hölder distortion. Dividing by $\int_Q \rho_Q^{\text{raw}}$ gives (70); the factor ℓ_Q removes the cell scale.

The homogeneity contribution has a fixed polynomial cost, while the strip mass is exponentially summable. The holonomy and chart contributions are summed by the context moment, and the amplitude contribution by the marked face ledger. Generalized Hölder with slightly smaller exponents proves (71). For a signed amplitude q , write $q = (2\|q\|_\infty + q) - 2\|q\|_\infty$; multiplication by the positive masses restores the scaled norm and introduces no reciprocal small-mass factor. $\square$

Lemma 6.30 (Fractional nonstationary phase). *Let I be an interval, $a \in C^\alpha(I)$, and $\phi \in C^{1+\alpha}(I)$. On each component of the support of a , suppose ϕ' has one sign and $|\phi'| \geq \lambda > 0$; assume a vanishes at every relative component endpoint. Put $\Phi = \sum_C \int_C |\phi'(t)| dt$, the total phase length of the support.*

$$\left| \int_I e^{i\phi(t)} a(t) dt \right| \leq C_\alpha \left\{ \frac{\|a\|_\infty}{\lambda} + \frac{\Phi[a]_\alpha}{\lambda^{1+\alpha}} + \frac{\Phi\|a\|_\infty[\phi']_\alpha}{\lambda^{2+\alpha}} \right\}. \quad (72)$$

The estimate adds over a countable union of sign components without a component-count loss.

Proof. On one sign component use $y = \phi(t)$ and write $b(y) = a(\phi^{-1}y)/\phi'(\phi^{-1}y)$. Then

$$\|b\|_\infty \leq \|a\|_\infty/\lambda, \quad [b]_\alpha \leq [a]_\alpha \lambda^{-1-\alpha} + \|a\|_\infty [\phi']_\alpha \lambda^{-2-\alpha}.$$

For a shift $h = \pi$, the identity

$$2 \int e^{iy} b(y) dy = \int e^{iy} \{b(y) - b(y+h)\} dy + \text{endpoint terms}$$

gives $C(\|b\|_\infty + \Phi[b]_\alpha)$. The endpoint terms vanish at interior sign boundaries because a does; the two exterior terms give the displayed supremum term. Summing components replaces their total y -length by at most Φ , proving (72). $\square$

Lemma 6.31 (Rank-one product under double centering). *If $\eta_{\pm}$ are admissible dual collision currents and $R = \eta_{-} \otimes \eta_{+}$, then*

$$R(Q_{-}^m g \otimes Q_{+}^n f) = \eta_{-}(Q_{-}^m g) \eta_{+}(Q_{+}^n f).$$

Thus the two one-sided gaps give $C\tau^m \vartheta^n$.

Proof. This is the universal property of the projective tensor product followed by the two admissible dual bounds. Admissibility of the projected marginals is proved in Lemma 6.34 below. $\square$

Proposition 6.32 (Central balanced estimate with explicit budget). *Let Θ be the energy exponent and set*

$$A_{\text{frac}} = 2 + \alpha, \quad 0 < \delta < \frac{\alpha}{2A_{\text{frac}}}, \quad \beta_{\text{cen}} = \min\{\delta\Theta/2, \alpha - A_{\text{frac}}\delta\} > 0.$$

Then every balanced nonzero graph mode has saving $C\Xi^{-\beta_{\text{cen}}}$ times a fixed power of $\text{Mark}(Q)$. If the observable regularity satisfies $r_{\text{obs}} > 2 + \beta_{\text{cen}}$, the complete lattice sum is exponentially small in the two induced depths. Rank-one and regular-product blocks have the same product decay.

Proof. Take $r = \Xi^{-\delta}$. The energy theorem bounds the resonant mass by $C\Xi^{-\delta\Theta/2} + Ce^{-cN}$. On its complement use two smooth cutoffs in $\psi'_{\xi}/(\Xi r)$, one for each sign. They vanish on the relative component boundaries and, by Proposition 6.29, their amplitudes satisfy

$$\|a_{\pm}\|_{\infty} + [a_{\pm}]_{\alpha} \leq C \text{Mark}(Q) r^{-1}.$$

Furthermore

$$\lambda = c\Xi r, \quad \Phi \leq C\Xi, \quad [\psi'_{\xi}]_{\alpha} \leq C\Xi \text{Mark}(Q).$$

Lemma 6.30 therefore gives

$$|\mathcal{I}_Q(\xi)| \leq C \text{Mark}(Q)^2 \{\Xi^{-1} r^{-1} + \Xi^{-\alpha} r^{-(2+\alpha)}\} \leq C \text{Mark}(Q)^2 \Xi^{-\beta_{\text{cen}}}.$$

The density moment and generalized Hölder sum the cells.

For completeness, choose $0 < \epsilon < c_{\text{gap}}/(4C_{\text{obs}})$. Modes with physical frequency $\Xi \leq e^{\epsilon N}$ are treated by the centered base gap; their Fourier-norm and counting loss is $e^{C_{\text{obs}}\epsilon N}$. In the range $e^{\epsilon N} < \Xi \leq e^{\chi+N}$, the preceding power saving is $e^{-\epsilon\beta_{\text{cen}}N}$, and Lemma 6.22 gives the buffer condition $q\delta\chi_{+} < \zeta$. Modes with $\Xi > e^{\chi+N}$ are summed from the $C^{r_{\text{obs}}}$ Fourier tail. The two-dimensional dyadic lattice sum converges because $r_{\text{obs}} > 2 + \beta_{\text{cen}}$. Zero and axis modes were removed or separated before localization; boundary-created axis modes use (50). Lemma 6.31 treats rank-one blocks, and the regular block is handled by Proposition 7.6. $\square$

6.11 Centered operator renewal and collision time

Lemma 6.33 (Local product for entrance/exit weights). *On the natural extension of the one-state Gibbs–Markov quotient, positive past and future cylinder weights are comparable to the product of their conditional one-sided weights, uniformly in the small parameter family. This assertion is used only for positive entrance/exit norms, not as a mixing estimate for signed observables.*

Proof. The concatenated Gibbs sum differs from the sum of its past and future pieces by a uniformly bounded distortion term. Exponentiation gives the comparison for cylinders and a monotone-class argument gives it for one-sided events; compare [22]. $\square$

Lemma 6.34 (Current and base spaces). *Let $F_{\pm} : Y_{\pm} \rightarrow Y_{\pm}$ be the one-state Gibbs–Markov bases with separation metrics d_{θ} . Set*

$$\mathcal{B}_{\pm} = C^{\theta}(Y_{\pm}), \quad \|h\|_{\mathcal{B}} = \|h\|_{\infty} + [h]_{\theta}, \quad \mathcal{B}_{\pm, w} = L^1(Y_{\pm}).$$

Let $\mathcal{C}_{\pm}$ be the signed projective completion of positive standard-pair currents:

$$\|\eta\|_{\mathcal{C}_{\pm}} = \inf_{\eta = \sum_j \epsilon_j \eta_j} \sum_j \eta_j(1) \{1 + Z_q(\eta_j) + [\log \rho_j]_{\alpha} + \text{Mark}(\eta_j)\}, \quad (73)$$

where $Z_q(\eta_j)$ is the normalized short-curve boundary functional and $\epsilon_j = \pm 1$. Its weak norm is total variation. Dynamically Hölder observables form the dual test space $\mathcal{C}'_{\pm}$.

Marginalization of a graph current and tensoring with a fixed smooth probability are bounded

$$\mathcal{G}_{\alpha} \longrightarrow \mathcal{C}_{\pm}, \quad \mathcal{C}_{-} \times \mathcal{C}_{+} \longrightarrow \mathcal{C}_{-} \widehat{\otimes}_{\pi} \mathcal{C}_{+}.$$

In particular the two projected marginals are admissible dual currents.

Proof. Disintegrate one positive graph piece over either endpoint. Proposition 6.29 controls the normalized density and its log–Hölder mark; the boundary functional is exactly the corresponding term in the graph projective norm. Integrating the unresolved coordinate and then taking the projective infimum proves the marginal bound. The tensor bound is the universal property of $\widehat{\otimes}_{\pi}$. $\square$

Lemma 6.35 (Current-side entrance/exit mapping). *All renewal operators act on currents:*

$$B_j^{\pm} : \mathcal{C}_{\pm} \rightarrow \mathcal{B}_{\pm}, \quad T_k^{\pm} : \mathcal{B}_{\pm} \rightarrow \mathcal{B}_{\pm}, \quad A_i^{\pm} : \mathcal{B}_{\pm} \rightarrow \mathcal{C}_{\pm}, \quad C_n^{\pm} : \mathcal{C}_{\pm} \rightarrow \mathcal{C}_{\pm}.$$

Under exponential return tails, bounded distortion, and the weighted one-step/growth estimates,

$$\|A_i^{\pm}\| + \|B_i^{\pm}\| + \|C_i^{\pm}\| \leq C e^{-ci} \quad (74)$$

in the strong norms, and the analogous strong-to-weak bounds hold. Moreover

$$B_j^{-} \widehat{\otimes} B_t^{+} : \mathcal{C}_{-} \widehat{\otimes}_{\pi} \mathcal{C}_{+} \rightarrow \mathcal{B}_{-} \widehat{\otimes}_{\pi} \mathcal{B}_{+}, \quad (75)$$

$$A_i^{-} \widehat{\otimes} A_s^{+} : \mathcal{B}_{-} \widehat{\otimes}_{\pi} \mathcal{B}_{+} \rightarrow \mathcal{C}_{-} \widehat{\otimes}_{\pi} \mathcal{C}_{+} \quad (76)$$

with norm $C e^{-c(i+j+s+t)}$.

Proof. For a positive standard pair, cut at the physical singularity and homogeneity curves before the first base visit. On a returned descendant W' ,

$$(B_j \eta)|_Y = \sum_{\tau_Y(W')=j} \{(\rho J_{W'}^{-1}) \circ h_{W'}\} dm_Y.$$

Full image puts every summand on the same base. If x, y have separation time $s(x, y)$, pair their inverse histories until the first separation. Distortion gives

$$|B_j \eta(x) - B_j \eta(y)| \leq C d_{\theta}(x, y) \sum_{\tau_Y(W')=j} \eta(W') \{1 + Z_q(W') + [\log \rho_{W'}]_{\alpha} + \text{Mark}(W')\}.$$

The growth inequality sums the braces and the exponential return tail gives e^{-cj} . This proves the B_j bound, including differences across countably many full branches. Push-forward gives A_i ; the same calculation restricted to descendants with no visit gives C_i . Positive-projective decomposition proves the signed bounds. Tensor bounds follow by the defining projective norm. $\square$

Lemma 6.36 (Base spectral disk). *For the aperiodic one-state Gibbs–Markov base with exponential return moment, $R_\pm(z) = \sum_{n \geq 1} z^n R_{\pm,n}$ acts analytically $\mathcal{B}_\pm \rightarrow \mathcal{B}_\pm$ on $|z| < 1 + \epsilon_0$, satisfies a uniform Doeblin–Fortet inequality into $\mathcal{B}_{\pm,w}$, and*

$$1 \notin \sigma(R_\pm(z)) \quad (0 < |z - 1| \leq \epsilon_1, |z| \leq 1 + \epsilon_1).$$

At $z = 1$, 1 is simple with residue $\Pi_\pm = \Pi_B^\pm$. Consequently

$$T_\pm(z) = (I - R_\pm(z))^{-1} = \frac{\Pi_\pm}{1 - z} + H_\pm(z)$$

with $H_\pm$ analytic on $|z| < 1 + \epsilon_1$, and $\|T_{\pm,n} - \Pi_\pm\|_{\mathcal{B} \rightarrow \mathcal{B}_w} \leq Ce^{-cn}$.

Proof. The branch formula and exponential return moment give operator-norm normal convergence. Bounded distortion gives the Doeblin–Fortet inequality and compact embedding into the weak space. Quasi-compactness follows. If $R(z)h = h$ for $|z| = 1$, equality in the triangle inequality forces the phase z^{R_a} to be constant on every return atom supporting h . Aperiodicity makes this possible only at $z = 1$; simplicity follows from mixing of the full-branch base. Analytic perturbation and compactness of the remaining annulus shrink ϵ_1 so that no other pole occurs. Cauchy’s estimate gives the coefficient bound. $\square$

Lemma 6.37 (Current-side tensor insertion). *Let $\mathcal{J} \in \mathcal{C}_- \widehat{\otimes}_\pi \mathcal{C}_+$ be globally double centered and put*

$$\mathcal{J}_{j,t}^B = (B_j^- \widehat{\otimes} B_t^+) \mathcal{J}.$$

Then

$$\|(E_k^- \widehat{\otimes} E_\ell^+) \mathcal{J}_{j,t}^B\|_w \leq CM e^{-c-k-c_+\ell-c(j+t)}. \quad (77)$$

For the graph summand the same entrance map preserves the density, slope, and boundary marks used in Proposition 6.32; for rank-one and regular summands it preserves their projective tensor norms. Applying $A_i^- \widehat{\otimes} A_s^+$ gains $e^{-c(i+s)}$.

Proof. Use Lemma 6.35 and the two base spectral gaps. The branch proof there transports exactly the density and boundary marks in the central phase estimate, so no unrecorded amplitude norm appears at entrance or exit. Apply the projective tensor property and then the exit bounds. Observables enter only in the final dual pairing. $\square$

Lemma 6.38 (Current-side centered tensor renewal). *On collision currents,*

$$\mathcal{P}_\pm^m = C_m^\pm + \sum_{i+k+j=m} A_i^\pm T_k^\pm B_j^\pm. \quad (78)$$

*Set $Q_\pm^{*m} = \mathcal{P}_\pm^m - \Pi_{\text{coll}}^\pm$, $E_k^\pm = T_k^\pm - \Pi_\pm$, and*

$$D_m^\pm = C_m^\pm + \sum_{i+k+j=m} A_i^\pm \Pi_\pm B_j^\pm - \Pi_{\text{coll}}^\pm. \quad (79)$$

Then

$$Q_\pm^{*m} = D_m^\pm + \sum_{i+k+j=m} A_i^\pm E_k^\pm B_j^\pm, \quad (80)$$

and $\|D_m^\pm\| + \|E_m^\pm\| \leq Ce^{-cm}$. Hence

$$|\langle (Q_-^{*m} \widehat{\otimes} Q_+^{*n}) \mathcal{J}, g \otimes f \rangle| \leq CM\tau^m \vartheta^n \|g\| \|f\|. \quad (81)$$

Proof. Lemma 6.36 gives the E_m tail. The residue identity $A_\pm(1)\Pi_\pm B_\pm(1) = \Pi_{\text{coll}}^\pm$ makes

$$D_\pm(z) = C_\pm(z) + \frac{A_\pm(z)\Pi_\pm B_\pm(z) - \Pi_{\text{coll}}^\pm}{1-z}$$

analytic on the same smaller disk; Cauchy's estimate gives the D_m tail. Tensor (80). The E - E contribution uses Lemma 6.37; mixed terms use one D tail and one centered gap; the D - D term uses their product. The coefficient sums are products of exponentially summable convolutions. $\square$

Theorem 6.39 (Induced Fourier estimate and centered renewal lift). *Assume Proposition 6.32, the off-diagonal estimates (22)–(23), and*

$$\sup_x \sum_a p_a(x) e^{\epsilon R_a} < \infty \quad (82)$$

in both orientations. Assume the aperiodic base spectral hypotheses of Lemma 6.38 and the weighted one-step expansion used in Lemma 6.35. Then

$$|B_{M,N}^{\text{cen}}(g, f)| \leq Ce^{-c(M+N)} \|g\|_{C^r} \|f\|_{C^r}. \quad (83)$$

If the full first/last visit decomposition includes the no-return remainder and aggregate projection correction as in (78)–(80), then

$$|B_{m,n}^{\text{cen}}(g, f)| \leq C\tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r}. \quad (84)$$

Proof. The three-region lemma combines the balanced estimate with the two one-sided coupling estimates. The low/middle/high physical-frequency split in Proposition 6.32, together with its total-time buffer ledger, gives (83). The exponential return moment makes the operator series analytic; aperiodicity and the base spectral decomposition give the E_m, D_m tails of Lemma 6.38. Lemmas 6.35 and 6.37 show coefficient by coefficient that the entrance and exit maps preserve the projective tensor norm. Apply the current-side identity of Lemma 6.38, evaluate on $g \otimes f$, and sum the marked amplitudes. $\square$

Lemma 6.40 (Negative short-curve moment). *For sufficiently small $q > 0$, every forward or reverse homogeneous cut satisfies*

$$\sum_{W' \in \mathcal{P}(W)} J_{W'} |W'|^{-q} \leq \vartheta_q |W|^{-q} + C_q J_W, \quad \vartheta_q < 1. \quad (85)$$

Consequently, for a positive standard family $\mathcal{G} = \{(W, \rho_W), p_W\}$,

$$Z_q(F^n \mathcal{G}) \leq \vartheta_q^n Z_q(\mathcal{G}) + C_q, \quad Z_q(\mathcal{G}) = \sum_W p_W |W|^{-q}.$$

Proof. Split descendants into those longer and shorter than a fixed fraction of $|W|$. Uniform cone expansion gives a factor Λ^{-q} on the first class. On the short class, group pieces by the singularity or homogeneity boundary which creates them. Linear complexity, bounded distortion, and the one-step expansion give $Cq + O(q^2)$ after raising lengths to $-q$. Choose q so that $\Lambda^{-q} + Cq + O(q^2) < 1$. Iteration and summation of the affine recursion give the family estimate. The reverse statement follows from reversibility with the same homogeneous atlas. $\square$

Proposition 6.41 (Radial Z_q moment and recovery). *Let $\Gamma = \{G(r, \varphi) = 0\}$, $|\nabla G| \geq c_G$, and cover it by the two implicit charts*

$$\mathfrak{C}_r = \{|\partial_\varphi G| \geq c_G/2\}, \quad \mathfrak{C}_\varphi = \{|\partial_r G| \geq c_G/2\}.$$

Relative to arclength,

$$\rho_\Gamma = \frac{|w_\Gamma|}{|\nabla G|} J_{\text{imp}} J_{\text{hol}}.$$

After the forward/reverse cone partition and positive decomposition, let $\mathcal{G}_\lambda^\pm$ be the two recovery families. For some $\epsilon_Z > 0$,

$$\sum_\lambda M_\lambda [(1 + Z_q(\mathcal{G}_\lambda^-))(1 + Z_q(\mathcal{G}_\lambda^+))]^{\epsilon_Z} < \infty. \quad (86)$$

Their recovery times satisfy

$$R_\lambda^\pm \leq a_1 \log(1 + Z_q(\mathcal{G}_\lambda^\pm)) + a_2, \quad (87)$$

and hence

$$|B_\lambda(m, n)| \leq C M_\lambda \tau^{(m-R_\lambda^-)_+} \vartheta^{(n-R_\lambda^+)_+}. \quad (88)$$

Proof. The two charts cover the regular face without division by a vanishing coordinate derivative. A smooth cone partition assigns each tangent to a forward or reverse recovery tree. On homogeneity strip b , the coarea flux is $M_b \leq C2^{-b}$, while the combined impact, holonomy, density, and chart cost is at most $2^{c_{\text{mark}}b}$ with $c_{\text{mark}} < 1$. Every component created by the initial finite face cut has square-root-chart length at least $c2^{-\kappa b}$; no such claim is made after recursive cuts. Thus, for the q of Lemma 6.40,

$$\sum_{\lambda: \text{rank}=b} M_\lambda Z_q(\mathcal{G}_\lambda^\pm)^{\epsilon_Z} \leq C2^{-[1-c_{\text{mark}}-\kappa q\epsilon_Z]b}. \quad (89)$$

Indeed, pieces are grouped by their creating boundary; their total $|W|^{-q}$ -weighted mass is exactly the left side of (85). Recursive cuts are controlled by iterating that affine recursion, rather than by a lower bound on individual lengths. Choose $q\epsilon_Z < (1 - c_{\text{mark}})/\kappa$ so that the exponent in (89) is positive and sum b . Generalized Hölder combines the two time directions, proving (86).

Canestrari's irregular-family recovery estimate gives (87). The positive/negative decomposition $q = (2\|q\|_\infty + q) - 2\|q\|_\infty$ preserves the mass-weighted log-Hölder and Z_q norms. Apply centered coupling to each recovered positive descendant and subtract. The $Z_q^{\epsilon_Z}$ moment absorbs all polynomial recovery windows, proving (88). $\square$

Proposition 6.42 (Four recovery quadrants). *Let a physical block be refined into masses $M_{b,j}$ with recovery times $R_{b,j}^-, R_{b,j}^+$. Under Proposition 6.41,*

$$|B_{s,b,j}(m, n)| \leq C M_{b,j} \tau^{(m-R_{b,j}^-)_+} \vartheta^{(n-R_{b,j}^+)_+}. \quad (90)$$

Proof. Before both recoveries use total variation. With only past recovery use (88); the future-only case is symmetric. After both recoveries apply Theorem 6.39 through the centered renewal identity. These four estimates are exactly (90). $\square$

7 Slope-energy and fractional-phase proof architecture

The proof now has six noncircular links.

- (F1) Quotient the exponential Young return on the regular magnet to the one-state countable $C^{1+\alpha}$ IFS of Proposition 6.10. Physical side labels remain in the signed amplitude fiber.
- (F2) Lemma 6.17 combines the exponential return tail with the maximal grazing rank and proves the joint moment needed for the countable quotient and the marked coarea amplitudes.
- (F3) The normal-cycle value $\mathfrak{A} = -1$, the exact holonomy series, the gate-hitting tail, and the mass-preserving resolving buffer give the conditional energy theorem and then Jensen. No countable QNL theorem or four-tree synchronization is inserted.
- (F4) Proposition 6.20 and smooth value-space cutoffs and Lemma 6.30 give the balanced Fourier estimate without curvature or componentwise phase partitions.
- (F5) Lemma 6.27 projects first in the extended current space. Only its graph block is then disintegrated; Lemma 6.28 removes contracted modes. Lemmas 6.35 and 6.37, together with the base spectral gap, lift induced decay through Lemma 6.38, with the no-return and projection remainders retained, and Proposition 6.41 treats the mixed quadrants.
- (F6) At each return cutoff L , coarea currents converge at fixed depth without matching connected components. Finite-depth generic face separation and a scalar conditional marked moment give one joint exponential tail and remove L before recovery summation.

This order makes clear which external results are used. Leclerc’s Lemma 3.15 supplies the cohomological bracket and his Propositions 4.31–4.33 supply the four-point all-scale polynomial escape. The final oscillatory estimate is the elementary fractional phase lemma proved above; LPS is retained only as background for the quotient route. Bertozzi’s triple inducing supplies a useful model for the one-state gate architecture, not the slope conclusion. The positive entrance/exit comparison is the symbolic local product argument of [22]. The one-step expansion, linear singularity complexity, and uniform sequential billiard constants used by the master atlas are summarized in [2, 3].

The route deliberately does not use the recent curvilinear two-ends theorem or L^2 flattening: their nonlinear applications require C^2 or higher geometry, whereas the billiard quotient available here is only $C^{1+\alpha}$. It also does not infer a product tower tail from two marginal Chernoff bounds: productization is the aggregate centered renewal identity (80) followed by a two-variable convolution.

7.1 Execution audit

The central analytic chain is now represented by explicit propositions. The remaining geometric inputs are the exponential Young return, uniform proper-family recovery, persistence of the clean normal basic set, and the finite-depth parameter continuation. The exact gate interface is Lemma 6.18, Proposition 6.19, and Lemma 6.22; the slope-energy theorem is Theorem 6.24; the balanced phase is Proposition 6.32; and collision time is Theorem 6.39. Recovery is applied only after these estimates, through Propositions 6.41 and 6.42 and then Proposition 4.1.

7.2 A quantitative smoothing test

There is one falsifiable way to obtain (17). Replace a face delta in block b by a normalized collar of width ϵ . Suppose the regularized bridge and the collar error satisfy

$$\|K_{b,\epsilon}\|_{X_- \rightarrow Y_+} \leq CM_b \epsilon^{-p}, \quad (91)$$

$$|((K_{V,b} - K_{b,\epsilon})Q^m g)(\mathcal{U}_0^n f)| \leq CM_b \epsilon^\alpha \Lambda_-^m \Lambda_+^n \|g\| \|f\|_{C^1}. \quad (92)$$

If the regularized dynamics gives rates $\tau_0^m \vartheta_0^n$, choosing $\epsilon = e^{-a(m+n)}$ yields CM2 provided one can choose $a > 0$ so that

$$\tau_0 e^{ap} < 1, \quad \vartheta_0 e^{ap} < 1, \quad \Lambda_- e^{-a\alpha} < 1, \quad qquad \Lambda_+ e^{-a\alpha} < 1. \quad (93)$$

Equivalently, a genuine spectral/bunching margin is required. For billiards, unbounded grazing derivatives make (92) the difficult part. The current flux estimate $M_b \lesssim 2^{-b}$ and recovery bound $R_b \lesssim b$ control the sum over b , but do not supply the normal Hölder estimate in (92). Thus collar smoothing is a testable fallback, not a completed proof.

Proposition 7.1 (Uniform difference-quotient closure). *Suppose either (15)–(16), or the shifted estimate (11) and the summability condition on the right of (12), hold uniformly with $K_{V,b}$ replaced by the assembled block of $s^{-1}(P_s - P_0)$. If each block converges weakly to $K_{V,b}$, then the parameter difference quotient of the prescribed susceptibility has a summable majorant and converges term by term to the double series generated by K_V .*

Proof. In the first case apply Theorem 3.2; in the second apply Proposition 4.1. Either hypothesis gives an integrable majorant after summing over b, m, n . Weak convergence on every block identifies the pointwise limit; dominated convergence completes the argument. $\square$

For the moving table, the uniform hypothesis in this proposition can itself be reduced to four checkable pieces.

Lemma 7.2 (Parameter derivatives of contracting return words). *Let $h_{j,s}$ be the inverse one-step branches in a fixed compact regular atlas and assume, uniformly in j and $|s| < s_0$,*

$$\|Dh_{j,s}\|_\infty \leq \lambda < 1, \quad \|\partial_s h_{j,s}\|_{C^1} + \|\partial_s \log |h'_{j,s}|\|_\infty \leq A.$$

For a clean word $a = (a_1, \dots, a_R)$ let $h_{a,s} = h_{a_1,s} \circ \dots \circ h_{a_R,s}$. Then

$$\|\partial_s h_{a,s}\|_\infty \leq \frac{A}{1-\lambda}, \quad \|\partial_s \log |h'_{a,s}|\|_\infty \leq C(1+R). \quad (94)$$

The same linear-in- R bound holds for products of uniformly C^1 Jacobian, stable-holonomy, and coarea factors. Hence an exponential return moment dominates the difference quotients of all clean marked amplitudes and their return tail.

Proof. Differentiate the composition. The contribution created at position k is multiplied on the left by at least $k-1$ contracting derivatives, so its C^0 norm is bounded by $A\lambda^{k-1}$; summation gives the first bound. The logarithmic derivative of a composition is a sum of R one-step logarithmic derivatives. Differentiating it gives a uniformly bounded direct term at every position and orbit-displacement terms controlled by the first bound and uniform C^2 distortion. This gives $C(1+R)$. The product statement follows after taking logarithms. Finally $R^p e^{-\epsilon R}$ is summable for every fixed p , so the exponential return moment absorbs all these factors. $\square$

This lemma shows that long return words do not create an exponential parameter-derivative loss on the inverse-branch side. The remaining issue is not the clean word calculus but the common refinement of words that approach or cross a moving singularity; those pieces must be charged to the already constructed face/recovery tree rather than differentiated as clean branches.

That singular refinement is most naturally made before fixing the parameter.

Lemma 7.3 (Finite-depth generic face separation). *Let $\mathbf{p}$ denote the centers and radii of a labelled circular table and let $\mathbf{v}$ be the radial deformation direction. Fix collision depth L and grazing-rank cutoff H . The active face equations $G_\lambda(\mathbf{p}, x) = 0$ form a finite analytic family. If $\lambda \neq \lambda'$ are not the same grouped physical preimage with the same multiplicity, there is a table-parameter direction $e_{\lambda\lambda'}$ such that*

$$\partial_{e_{\lambda\lambda'}} G_\lambda \not\equiv \partial_{e_{\lambda\lambda'}} G_{\lambda'} \quad \text{on a common regular germ.} \quad (95)$$

Consequently the common-factor resultant and the first-jet transversality determinant are not both identically zero. Their exceptional set $\mathcal{E}_{L,H}$ is a proper real-analytic subset. Outside it, distinct faces meet transversely in zero-dimensional strata.

The complement of $\bigcup_{L,H \geq 1} \mathcal{E}_{L,H}$, together with the analogous exceptional set of deformation directions $\mathbf{v}$, is residual and has full Lebesgue measure.

Proof. There are finitely many labelled itineraries at fixed L, H . For two distinct labels, choose the earliest collision at which their obstacle, side, or tangency record differs. If the obstacle differs, translate the center of the obstacle used by λ in the normal direction at that collision; the derivative of G_λ is nonzero by the impact transversality, while the earlier jet of $G_{\lambda'}$ is unchanged. If the obstacle agrees but the side or tangency multiplicity differs, vary its radius; the first nonmatching normal jet separates the two equations. This proves (95). The identical response is possible only when the physical preimage and multiplicity agree, in which case assembly groups the faces.

Elimination of the phase variables gives the common-factor resultant; adjoining the determinant of the two phase gradients detects nontransverse intersection. The separating direction shows that the analytic pair is not identically zero, so its zero set is proper and has Lebesgue measure zero. The finite union gives $\mathcal{E}_{L,H}$. Countable intersection over L, H gives the last assertion. $\square$

Proposition 7.4 (Finite-depth coarea and conditional-label moment). *Assume*

$$|\nabla_x G_\lambda(s, x)| \geq c_G > 0. \quad (96)$$

At fixed L, H , every moving indicator has the coarea derivative

$$\frac{\mathbf{1}_{\{G_\lambda(s, \cdot) > 0\}} - \mathbf{1}_{\{G_\lambda(0, \cdot) > 0\}}}{s} = \int_0^1 \delta(G_\lambda(\theta s, \cdot)) \partial_s G_\lambda(\theta s, \cdot) d\theta, \quad (97)$$

and these currents converge weakly.

Let $\mathcal{F}_k$ be the filtration which successively exposes return, face, homogeneity, recursive-cut, and forward/reverse recovery descendants. Write $\Delta \mathfrak{D}_k$ and ΔK_k for the new cost and mark at node k . The return, growth, context, and radial moment lemmas give numbers $\epsilon_0, p_0, c_0 > 0$ such that

$$\mathbb{E} \left[e^{\epsilon_0 \Delta \mathfrak{D}_k} (1 + \Delta K_k)^{p_0} \mid \mathcal{F}_{k-1} \right] \leq e^{c_0} \quad (98)$$

at every nonterminal node. The exponential return exponent is larger than the accumulated node cost c_0 per collision. Hence, for some $\epsilon_*, p > 1$,

$$\sup_{|s| < s_0} \sum_{\lambda} K_{\lambda}^p e^{\epsilon_* \mathfrak{D}_{\lambda}} \mu_s^u(J_{\lambda,s}) < \infty, \quad (99)$$

where

$$\mathfrak{D}_{\lambda} = R_{\lambda} + B_{\lambda} + C_{\text{cut},\lambda} + R_{\lambda}^{-} + R_{\lambda}^{+}, \quad K_{\lambda} = 1 + Z_q(\mathcal{G}_{\lambda}^{-}) + Z_q(\mathcal{G}_{\lambda}^{+}) + \mathfrak{b}_{\lambda} + |\partial_s \log J_{\lambda}|.$$

Consequently

$$\sup_{|s| < s_0} \sum_{\lambda: \max\{R_{\lambda}, B_{\lambda}, C_{\text{cut},\lambda}, R_{\lambda}^{-}, R_{\lambda}^{+}\} > L} \|w_{\lambda,s}\|_{\mathcal{A}^{\alpha}} \mu_s^u(J_{\lambda,s}) \leq C e^{-cL}. \quad (100)$$

Proof. Formula (97) is the fundamental theorem of calculus for $H(G_{\lambda})$, followed by coarea. Lemma 7.3 leaves finitely many transverse regular strata; their zero-dimensional intersections have no one-dimensional coarea mass. This proves fixed- (L, H) convergence.

For (98), return and context nodes use their exponential moments, homogeneity nodes use the strip flux, cut nodes use Lemma 6.40, and recovery nodes use (86) and the logarithmic recovery bound. Parameter derivatives have polynomial word cost and are absorbed by a smaller return exponent. Iterate conditional expectation until the terminal label. A history of R collisions contributes at most $e^{c_0 R}$; shrinking ϵ_0 leaves a positive margin in the return exponential moment. Generalized Hölder combines the fixed powers and proves (99). Markov's inequality gives (100), uniformly removing both collision and grazing cutoffs after the parameter limit. $\square$

This construction is why a differentiable parameter family of Young towers is unnecessary. The dynamical return alphabet is fixed on clean master cells, while every atom which changes its itinerary is represented by the physical coarea current that the response formula already contains.

The clean cohomological bracket and the master refinement can now be put on one ledger. The point of the next proposition is not to create a second symbolic dynamics. It records, on the amplitude side, which already existing full branches and recovery descendants contain a physical face piece.

Proposition 7.5 (Global labelled bi-return ledger). *Shrink s_0 so that the regular magnet, the clean normal basic set, and its fixed homogeneous charts persist for $|s| < s_0$. Use the parameter–state master refinement of Proposition 7.4, the marked Young quotient of Proposition 6.10, and the common past/future recovery tree of Corollary 4.4. Then, modulo zero face measure, the complete physical first defect has a disjoint signed decomposition indexed by labels*

$$\lambda = (\chi; a, k, \omega, \ell; b, j^{-}, j^{+}; c^{-}, g, m, c^{+}; \nu). \quad (101)$$

Here χ is a fixed physical chart, (a, k, ω, ℓ) records the clean return word, insertion time, finite physical cut history, and one-sided mark, b is the homogeneity rank, (j^{-}, j^{+}) is the common past/future recovery descendant, and (c^{-}, g, m, c^{+}) records the clean entrance context, the gate visit, the mass-preserving resolving-buffer depth, and the clean exit context, while ν is the measurable component index created at the last finite cut. Thus ν is inherited recursively and is not reordered when endpoints move with the parameter. Inactive entries are given a null symbol; in particular $b = 0$ and $j^{\pm} = \emptyset$ on the compact central core.

For every label and every parameter slice on which it is nonempty:

- (L1) *all signs of the physical inequalities $G_{a,r}$ are fixed in the relative interior, so c^- and c^+ are homogeneous clean words and Lemma 6.18 and Proposition 6.19 apply on that same atom;*
- (L2) *the sets selected by ℓ, b, j^-, j^+ and by the master signs are amplitude restrictions only. The quotient alphabet remains the uncut full-branch alphabet $\{h_a\}$ (or its full-branch triple acceleration). Put $\text{Mark}(\lambda) = \text{Mark}(Q_\lambda)$ as in (69); the conditional energy estimate is run before multiplication by the restricted signed amplitude. We also abbreviate $Z_{b,j^\pm}^\pm = Z_q(G_{b,j^\pm}^\pm)$ for the two recovered standard families;*
- (L3) *one may choose a single constant ledger, independent of λ and s ,*

$$\mathfrak{C} = (C_{\text{at}}, c_{\text{tr}}^{-1}, \lambda_*^{-1}, D_*, H_*, E_{\text{ret}}, A_{\text{mark}}, C_{\text{blow}}, p_G^{-1}, q, C_{\text{br}}, C_{\text{cur}}, C_{\text{ins}}, C_{\text{ren}}, C_{\text{stop}}, C_{\text{tail}}, C_{\text{par}}, C_{\text{rec}}, \tau^{-1}, \vartheta^{-1}), \quad (102)$$

controlling respectively the finite atlas, physical transversality, quotient contraction, distortion and holonomy, the exponential return moment, the marked-amplitude moment, the polynomial blow-up, gate-hitting probability and resolving buffer, the cohomological bracket tail, the aggregate centered renewal convolution, the extended-current projection, bilinear entrance/exit insertion, centered tensor renewal, radial Z -moment tree, scalar marked tail, parameter derivatives, and the two-sided recovery estimate;

- (L4) *for some $\epsilon' > 0$ the clean and grazing parts obey the joint summability bound*

$$\begin{aligned} & \sum_{\lambda: b=0} e^{\epsilon'(|c^-|+|c^+|)} \|w_{\lambda,s}\|_{\mathcal{A}^\alpha(J_{\lambda,s})} \mu_s^u(J_{\lambda,s}) (1 + [S_\lambda]_{C^\alpha})^{p_F} \text{Mark}(\lambda)^{p_d} \\ & + \sum_{b \geq 1, j^-, j^+} M_{b,j^-,j^+} (1 + R_{b,j^-}^-)(1 + R_{b,j^+}^+) [(1 + Z_{b,j^-}^-)(1 + Z_{b,j^+}^+)]^{\epsilon_Z} \leq C(\mathfrak{C}), \end{aligned} \quad (103)$$

uniformly in s .

On every clean labelled atom the coarea identity (97) and the cohomological bracket identity (41) therefore hold simultaneously with the same constants. On the complementary atoms the recovery ledger (86) holds with the same constants; the decay is derived later by Proposition 6.42.

Proof. Start with a finite master cut-history cell and intersect it successively with a physical face, a common recovery descendant, and a triple-induced cylinder. These operations commute modulo face-null boundaries: master and cylinder cuts are restrictions in the relative interior, while coincident artificial faces have already cancelled with opposite signs. Lemma 7.3 groups coincident physical faces and makes every remaining finite-depth intersection face-null, so no new first-order atom is created. The orbit itinerary, finite cut history, and inherited component index on a surviving cell uniquely give (101); conversely the label specifies that cell up to its null boundary. This proves disjointness and exhausts the complete first defect.

The cohomological bracket formula is a wordwise identity along a homogeneous word, hence it survives every subsequent restriction. The same is true of the distributional coarea formula on each parameter slice. Importantly, no master or one-sided cut is declared to be a new inverse branch. Thus the full-image property and the light-tail moment are those of the original quotient. The conditional slope energy is first applied to that Gibbs

conditional and Jensen is taken after branch averaging; the restricted physical weight is then inserted by Lemma 6.9. Consequently a thin master piece is never used as a new conditional; light derivative cylinders are charged by (54), not assigned a false uniform lower mass.

Choose the chart, transversality, hyperbolicity, distortion, and holonomy constants on the compact persistent regular atlas. Choose the polynomial escape, gate-hitting, and buffer constants from Proposition 6.19, Lemma 6.22, and Theorem 6.24. Lemma 6.18 gives the word-independent bracket-tail constant C_{br} , and Lemma 6.38 gives C_{ren} ; compactness in s makes them uniform. Restriction cannot increase the scale-normalized norm (31). Parameter differentiation costs at most a polynomial in the word length by Lemma 7.2, and the exponential return and context moments absorb every such factor. This fixes (102) before the labels are summed.

For $b = 0$, additivity of total variation on disjoint restrictions and Proposition 6.11 give the first sum in (103); Lemma 6.16 gives the joint exponential moment of the two context lengths without independence, while (44), Proposition 6.29, and generalized Hölder absorb the displayed slope and density marks. A return word contributes linearly many physical insertion times; recursive singularity components are summed by weighted one-step expansion. For $b \geq 1$, refinement does not increase the sum of the descendant total variations; Lemma 6.40 and Proposition 6.41 give the Z_q and recovery factors in the second sum. This proves the uniform joint bound. Every atom is either clean after its recorded recovery times, when both identities apply, or lies in its recorded recovery window, where the common recovery tree supplies the second term of (103). Only after the independent central estimate is applied does Proposition 4.1 sum the resulting shifted decay. $\square$

Proposition 7.6 (Regular defect in the tensor current space). *Let K_{reg} be the assembled two-dimensional Jacobian and density derivative of the first defect. On each homogeneous chart its kernel has scaled C^α norm bounded by the regular marked ledger. After global double centering it defines an element of $\mathcal{C}_- \hat{\otimes}_\pi \mathcal{C}_+$ and*

$$\|K_{\text{reg}}\|_{\mathcal{C}_- \hat{\otimes}_\pi \mathcal{C}_+} \leq C \sum_\lambda \|k_\lambda\|_{C_{\text{sc}}^\alpha} \mu^-(U_\lambda^-) \mu^+(U_\lambda^+) < \infty. \quad (104)$$

Moreover

$$|\langle (Q_-^{*m} \hat{\otimes} Q_+^{*n}) K_{\text{reg}}, g \otimes f \rangle| \leq C \tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r}. \quad (105)$$

Proof. Write each chart kernel as a projective sum of scaled C^α rectangles. Multiplication by a scaled C^α regular density is bounded on the positive-projective collision space, and its two marginals are admissible by Lemma 6.34. The global ledger sums the homogeneous chart norms, proving (104). Apply the entrance/exit tensor maps, the base E – E spectral gap, and the current-side renewal convolution. The D – E , E – D , and D – D terms use the exponential D tail exactly as in Lemma 6.38. This proves (105) without reducing a two-time estimate to a one-time correlation bound. $\square$

Theorem 7.7 (CM2 for a generic clean circular table). *Let the circular table and radial direction lie outside the countable generic exceptional set of Lemma 7.3. Assume a compact finite-horizon chart with:*

- (H1) *the standard one-state marked Young quotient with exponential return and context moments, bounded distortion, and aperiodicity;*
- (H2) *the positive and negative weighted one-step/growth inequalities;*
- (H3) *the persistent clean normal cycle with $\mathfrak{A} = -1$;*

(H4) *uniform face nondegeneracy and the assembled coarea/current ledger.*

Then the complete double-centered physical first defect satisfies

$$|\langle Q^n K_V Q^m g, f \rangle| \leq C \tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r}.$$

The assembled difference quotients have the same summable majorant and their CM2 series converges term by term as the table parameter tends to zero.

Proof. The exact endpoint bracket, physical stopping cover, legal-child recursion, and total-time buffer give Theorem 6.24. Proposition 6.29 and Lemma 6.30 give the complete balanced-mode estimate. Lemmas 6.34, 6.35, and 6.36 verify the current-side tensor renewal hypotheses, so Theorem 6.39 gives the clean central product estimate.

For the radial pieces, Lemma 6.40 and Proposition 6.41 give the two-sided recovery moment; Proposition 6.42 covers the four recovery quadrants and the recovery-window lemma sums them. Proposition 7.6 treats the regular two-dimensional part. Finally, Lemma 7.3 and Proposition 7.4 give fixed- (L, H) coarea convergence and a uniform marked tail. Take the parameter limit first and then remove L, H . No conclusion of CM2 is used as a hypothesis. $\square$

Proposition 7.8 (Uniform coarea-current majorant). *In a compact generic table chart satisfying (H1)–(H4), all constants in the energy, fractional-phase, tensor-renewal, radial-recovery, and parameter-tail estimates are uniform. Hence $s^{-1}(P_s - P_0)$ has a common summable CM2 majorant.*

Proof. Hyperbolicity, distortion, the clean gate, and single-face nondegeneracy are open on the fixed compact chart. The return/context moments and both one-step inequalities are uniform by (H1)–(H2). Therefore the stopping-cover constants, density marks, current norms, spectral disk, and negative short-curve exponent can be chosen once. At fixed L, H , the generic resultants stay nonzero after shrinking the chart around the base table. Proposition 7.4 supplies a uniform tail for leaving that finite atlas. Apply the already proved central and recovery estimates label by label, then sum the scalar marked majorant. Dominated convergence completes the parameter limit. $\square$

The construction does not require a differentiable parameter family of countable towers: finite-depth coarea convergence is taken before the parameter limit, and the collision/grazing cutoffs are removed by the conditional-label moment.

8 Global ledger and residual falsification checks

The central audit is represented by the exact holonomy series, the context-polynomial blow-up, the gate-hitting tail, and the mass-preserving resolving buffer. Lemma 6.22, Theorem 6.24, and Corollary 6.25 give the resonant physical slope estimate directly; Lemma 6.26 converts it to the derivative lower bound required by Lemma 6.30. Propositions 7.4, 6.41, and 6.42, together with the base spectral disk, close the remaining steps in the noncircular order “central decay, fibrewise recovery, summation.” The finite clean escape seed is supplied by Proposition 6.13, and the quotient weights and marked amplitudes are supplied by Propositions 6.10 and 6.11. Together these imply the local estimate

$$\left| \int_I g(T^{-m} \gamma(t)) f(T^n F \gamma(t)) w(t) dt \right| \leq C \tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r}. \quad (106)$$

using Proposition 6.23, Lemmas 6.22, 6.35, and 7.8, together with the weighted-transfer lemma. Proposition 7.5 now supplies the single global label map and proves that the constants in (41) agree on every fixed-depth parameter–state common refinement. A master restriction is an amplitude mark, not a new Gibbs conditional; light derivative cylinders are paid by (54).

The high-radial-block check is now an audit rather than a new mixing theorem: write the common past/future growth-tree refinement explicitly and match its weights to the fixed-atlas physical block bound used in Corollary 4.4. High grazing ranks then contribute only

$$\sum_b M_b(1+b)^2 < \infty,$$

and no blockwise Fourier constant is needed. A failure of this bookkeeping would invalidate the corollary, but it would not reopen the central slope calculation.

In parallel, one may still test the stronger mixed-cylinder condition (17). Success would imply CM2 immediately, but failure would only show that the graph current is non-nuclear; it would not refute (106).

The current status is therefore precise: future recovery is available, complete centering is performed by global projections, the prescribed source is not a coboundary, and a fixed circular normal cycle gives an exact clean escape seed. The principal path no longer contains a countable QNL theorem. Proposition 6.19, Lemma 6.22, and Theorem 6.24 import all-scale escape on each physical branch pair, absorb context jets into the polynomial target, and charge light cylinders without four-tree synchronization. The global label/constant audit is closed by Proposition 7.5; Theorem 7.7 gives the assembled CM2 statement and its parameter-uniform dominated-convergence version. The two residual checks above are independent falsification routes for, respectively, the stronger nuclear criterion and the already isolated recovery constants; neither is a missing implication in the CM2 assembly.

References

- [1] M. Demers and C. Liverani, *Projective cones for sequential dispersing billiards*, Commun. Math. Phys. 401 (2023), arXiv:2104.06947.
- [2] M. Demers and C. Liverani, *Recent progress in the application of transfer operators to dispersing billiards*, arXiv:2606.10155 (2026).
- [3] M. Demers and C. Liverani, *Central limit theorem for sequential dynamical systems*, arXiv:2502.07765v2 (2025).
- [4] M. Stenlund, L.-S. Young and H.-K. Zhang, *Dispersing billiards with moving scatterers*, arXiv:1210.0011.
- [5] J. Leppänen and M. Stenlund, *A note on the finite-dimensional distributions of dispersing billiard processes*, arXiv:1702.01461.
- [6] C. Wormell, *Conditional mixing in deterministic chaos*, arXiv:2206.09291; and *On convergence of linear response formulae in piecewise hyperbolic maps*, arXiv:2206.09292.
- [7] S. Baker and A. Banaji, *Polynomial Fourier decay for fractal measures and their pushforwards*, Math. Ann. 392 (2025), 209–261, arXiv:2401.01241.

- [8] S. Baker, O. Khalil and T. Sahlsten, *Fourier decay from L^2 -flattening*, arXiv:2407.16699.
- [9] V. Climenhaga and J. Day, *Every finite horizon Sinai billiard map has a unique measure of maximal entropy*, arXiv:2604.25881.
- [10] P. Bálint, P. Nándori, D. Szász and I. P. Tóth, *Equidistribution for standard pairs in planar dispersing billiard flows*, Ann. Henri Poincaré 19 (2018), arXiv:1707.03068.
- [11] O. Butterley, G. Canestrari and R. Castorrini, *Discontinuities cause essential spectrum on surfaces*, Ann. Henri Poincaré (2024), arXiv:2306.00484.
- [12] G. Canestrari, *Linear response for Sinai billiards with small holes*, arXiv:2604.19671v2 (2026).
- [13] G. Leclerc, *Fourier decay of equilibrium states for bunched attractors*, arXiv:2301.10623.
- [14] G. Leclerc, *Fourier decay of equilibrium states and the Fibonacci Hamiltonian*, arXiv:2507.23731v2 (2026).
- [15] G. Leclerc, S. Paukkonen and T. Sahlsten, *Fourier decay in parabolic $C^{1+\alpha}$ systems with overlaps*, arXiv:2505.15468v3 (2025).
- [16] A. Algom, F. Rodriguez Hertz and Z. Wang, *Polynomial Fourier decay and a cocycle version of Dolgopyat’s method for self conformal measures*, arXiv:2306.01275v2 (2024).
- [17] Y. Lima, D. Obata and M. Poletti, *Measures of maximal entropy for non-uniformly hyperbolic maps*, arXiv:2405.04676v2 (2025).
- [18] N. Chernov, *Decay of correlations and dispersing billiards*, J. Stat. Phys. 94 (1999), 513–556.
- [19] P. Teixeira, *On the conservative pasting lemma*, Ergodic Theory Dynam. Systems 40 (2020), 1402–1440, arXiv:1611.01694.
- [20] R. Rühr, *Pressure inequalities for Gibbs measures of countable Markov shifts*, arXiv:2012.13226v3.
- [21] N. Bertozzi, *Exponential mixing for hyperbolic flows on non-compact spaces*, arXiv:2603.11852v2 (2026).
- [22] V. Climenhaga, *Gibbs measures have local product structure*, arXiv:2310.17495v2 (2025).
- [23] S. Gouëzel, *Berry–Esseen theorem and local limit theorem for non uniformly expanding maps*, Ann. Inst. H. Poincaré Probab. Statist. 41 (2005), 997–1024, arXiv:math/0310391.
- [24] D. Galli, *A cohomological approach to Ruelle–Pollicott resonances and speed of mixing of Anosov diffeomorphisms*, arXiv:2405.17045.